

1.What is the composition of our respondents by age (group by 10 years like 10-20, 21-30, etc.), gender, caste and religion? What is the average family size and median age of family members of the respondents? Categorize by religion and if Hindu, by caste.

1.1. What is the composition of our respondents by age (group by 10 years like 10-20, 21-30, etc.), gender, caste and religion?

Query asked:

group the age with 10 years interval age groups than group by the table based on religion, caste, gender and age group

Generated SQL query:

```
SELECT
    Religion,
    Category_of_Caste,
    Gender,
    CONCAT(
        FLOOR(Age / 10) * 10,
        '-',
        FLOOR(Age / 10) * 10 + 9
    ) AS age_group,
    COUNT(*) AS respondent_count
FROM
    respondent
WHERE
    Age IS NOT NULL
GROUP BY
    Religion,
    Category_of_Caste,
    Gender,
    age_group
ORDER BY
    Religion,
    Category_of_Caste,
```

Gender,
age_group
Output:

Religion	Category_of_Caste	Gender	age_group	respondent_count
	OBC	Female	30-39	1
	SC	Female	50-59	1
Christian	OBC	Female	20-29	1
Christian	OBC	Female	30-39	1
Christian	SC	Female	Oct-19	1
Hindu		Female	Oct-19	1
Hindu		Female	30-39	1
Hindu		Female	40-49	1
Hindu		Male	20-29	1
Hindu	General	Male	60-69	1
Hindu	OBC	Female	30-39	1
Hindu	OBC	Female	40-49	1
Hindu	OBC	Male	40-49	2
Hindu	SC	Female	30-39	1
Hindu	SC	Female	40-49	1
Hindu	SC	Male	20-29	1
Hindu	SC	Male	60-69	1
Hindu	ST	Female	20-29	2
Hindu	ST	Female	30-39	1
Hindu	ST	Female	40-49	1
Hindu	ST	Male	30-39	1
Hindu	ST	Male	60-69	1
Muslim		Female	20-29	1
Muslim		Female	30-39	1
Muslim	General	Male	30-39	1
Prefer Not to Say	SC	Female	0-9	1
Prefer Not to Say	ST	Female	30-39	1

2. Same as above for educational qualifications. Make a crosstab of housing tenure (rent/own etc.) by highest educational qualification in the respondents' families?

```
2. --- SQL Agent Interactive Mode ---
3. Type your database question and press Enter. Type 'exit' or 'quit' to
   end the session.
4.
5. Query (or 'exit'): How many respondents live in their own house, on rent
   or any other arrangement? . Make a crosstab of housing tenure (rent/own
   etc.) by highest educational qualification in the respondents' families?
6.
7. --- Processing Query: How many respondents live in their own house, on
   rent or any other arrangement? . Make a crosstab of housing tenure
   (rent/own etc.) by highest educational qualification in the respondents'
   families? ---
8.
9.
10.
11.      > Entering new SQL Agent Executor chain...
12.      Thought: The user wants two pieces of information:
13.      1. The count of respondents by their housing tenure.
14.      2. A crosstab of housing tenure by the highest educational
       qualification within each respondent's family.
15.
16.      Let's break this down into manageable steps.
17.
18.      **Part 1: Count of respondents by housing tenure.**
19.      I need to query the `Respondent` table and count the occurrences of
       each value in the `Is_this_house_rented/_owned` column.
20.
21.      **Part 2: Crosstab of housing tenure by highest educational
       qualification in the respondents' families.**
22.      This requires a more complex query.
23.      First, I need to define an order for educational qualifications to
       determine the "highest".
24.      Education levels and their assigned ranks (higher rank means higher
       education):
25.      - Illiterate/Never attended school: 0
26.      - PrePrimary: 1
27.      - Primary (class 1 - 5): 2
28.      - Secondary (class 6 - 10): 3
29.      - Higher Secondary/ Intermediate (class 11 - 12): 4
30.      - Graduation: 5
31.      - Empty string/N/A: -1 (will be handled as a separate category or
       ignored if not relevant)
32.
33.      I will need to:
34.      1. Get the housing tenure for each respondent from the
       `Respondent` table.
35.      2. For each respondent's household (linked by `_uuid` in
       `Respondent` and `_submission_uuid` in `Household_members`), determine
       the highest educational qualification among all members, including the
       respondent. This will involve a `UNION ALL` between the respondent's
```

education and all household members' education, then finding the maximum rank.

36. 3. Join these two pieces of information and then pivot or group to create the crosstab.

37.

38. Let's start with Part 1.

39.

40. ****Step 1: Get the count of respondents by housing tenure.****

41. Action: `sql_db_query_checker`

42. Action Input: `SELECT "Is_this_house_rented/_owned", COUNT(*) FROM "Respondent" GROUP BY "Is_this_house_rented/_owned";SELECT "Is_this_house_rented/_owned", COUNT(*) FROM "Respondent" GROUP BY "Is_this_house_rented/_owned";Thought:The query for Part 1 seems correct. I will execute it.`

43.

44. Action: `sql_db_query`

45. Action Input: `SELECT "Is_this_house_rented/_owned", COUNT(*) FROM "Respondent" GROUP BY "Is_this_house_rented/_owned";[('Own', 10), ('Rent', 4), ('Squatting', 14), ('Temporarily staying as Guests in Relative/Friends place', 1)]Thought:I have successfully retrieved the counts for housing tenure. Now I need to tackle the crosstab part.`

46.

47. To create the crosstab of housing tenure by the highest educational qualification in the respondents' families, I need to:

48. 1. Define a ranking for educational qualifications for both `Respondent` and `Household\_members` tables.

49. 2. For each respondent, find their own highest education qualification.

50. 3. For each respondent's household, find the highest education qualification among all `Household\_members`.

51. 4. Combine the respondent's education and household members' education to find the overall highest education for the family.

52. 5. Join this with the housing tenure from the `Respondent` table.

53. 6. Pivot the results to create the crosstab.

54.

55. Let's define the education ranking:

56. - 'Illiterat/Never attended school' (Respondent) / 'Illiterate/Never_attended_school' (Household_members): 0

57. - 'PrePrimary' (Household_members): 1

58. - 'Primary (class 1 - 5)' (Respondent) / 'Primary_(class_1__5)' (Household_members): 2

59. - 'Secondary (class 6 - 10)' (Respondent) / 'Secondary_(class_6__10)' (Household_members): 3

60. - 'Higher Secondary/ Intermediate (class 11 - 12)' (Respondent) / 'Higher_Secondary/_Intermediate_(class_11__12)' (Household_members): 4

61. - 'Graduation' (Respondent) / 'Graduation' (Household_members): 5

62. - Empty string/N/A: -1 (will be handled as a separate category or ignored if not relevant)

63.

64. I will create a CTE for each table to assign a numeric rank to the education levels and then find the max rank per family.

65.

66. ****Step 2: Create a CTE for Respondent education ranks.****

67. ****Step 3: Create a CTE for Household_members education ranks.****

68. ****Step 4: Combine and find the highest education rank per family.****

69. ****Step 5: Join with housing tenure and pivot.****

70.

71. Let's construct the query for the crosstab. This will be a multi-step SQL query.

72.

```

73.     First, I'll create a CTE for the respondent's education with a
       numerical rank.
74.     Then, a CTE for household members' education with a numerical rank,
       linked to the respondent via `_submission_uuid`.
75.     Then, I'll combine these to find the maximum education rank for
       each household.
76.     Finally, I'll join with the housing tenure and group to create the
       crosstab.
77.
78.     ```sqlite
79.     WITH RespondentEducation AS (
80.         SELECT
81.             _uuid,
82.             "Is_this_house_rented/_owned",
83.             CASE "Highest_Education_Qualification"
84.                 WHEN 'Illiterat_/Never attended school' THEN 0
85.                 WHEN 'Primary (class 1 - 5)' THEN 2
86.                 WHEN 'Secondary (class 6 - 10)' THEN 3
87.                 WHEN 'Higher Secondary/_Intermediate (class 11 - 12)'
88.             THEN 4
89.                 WHEN 'Graduation' THEN 5
90.                 ELSE -1 -- Handle empty or unknown values
91.             END AS EducationRank
92.         FROM "Respondent"
93.     ),
94.     HouseholdMembersEducation AS (
95.         SELECT
96.             _submission_uuid,
97.             CASE "Highest_Education_Qualification"
98.                 WHEN 'Illiterate/Never attended_school' THEN 0
99.                 WHEN 'PrePrimary' THEN 1
100.                WHEN 'Primary (class_1_5)' THEN 2
101.                WHEN 'Secondary (class_6_10)' THEN 3
102.                WHEN 'Higher_Secondary/_Intermediate (class_11_12)'
103.            THEN 4
104.                WHEN 'Graduation' THEN 5
105.                ELSE -1 -- Handle empty or unknown values
106.            END AS EducationRank
107.        FROM "Household_members"
108.    ),
109.    FamilyHighestEducation AS (
110.        SELECT
111.            R._uuid,
112.            R."Is_this_house_rented/_owned",
113.            MAX(T.EducationRank) AS MaxEducationRank
114.        FROM RespondentEducation AS R
115.        LEFT JOIN (
116.            SELECT _uuid, EducationRank FROM RespondentEducation
117.            UNION ALL
118.            SELECT _submission_uuid AS _uuid, EducationRank FROM
119.            HouseholdMembersEducation
120.        ) AS T ON R._uuid = T._uuid
121.        GROUP BY R._uuid, R."Is_this_house_rented/_owned"
122.    )
123.    SELECT
124.        F."Is_this_house_rented/_owned",
125.        SUM(CASE WHEN F.MaxEducationRank = 0 THEN 1 ELSE 0 END) AS
126.        "Illiterate/Never attended school",
127.        SUM(CASE WHEN F.MaxEducationRank = 1 THEN 1 ELSE 0 END) AS
128.        "PrePrimary",

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124.         SUM(CASE WHEN F.MaxEducationRank = 2 THEN 1 ELSE 0 END) AS
        "Primary (class 1 - 5)",
125.         SUM(CASE WHEN F.MaxEducationRank = 3 THEN 1 ELSE 0 END) AS
        "Secondary (class 6 - 10)",
126.         SUM(CASE WHEN F.MaxEducationRank = 4 THEN 1 ELSE 0 END) AS
        "Higher Secondary/ Intermediate (class 11 - 12)",
127.         SUM(CASE WHEN F.MaxEducationRank = 5 THEN 1 ELSE 0 END) AS
        "Graduation",
128.         SUM(CASE WHEN F.MaxEducationRank = -1 THEN 1 ELSE 0 END) AS
        "Unknown/N/A"
129.     FROM FamilyHighestEducation AS F
130.     GROUP BY F."Is_this_house_rented/_owned";
131.     ``
132.
133.     Let's check this complex query.
134.     Action: sql_db_query_checker
135.     Action Input: WITH RespondentEducation AS (
136.         SELECT
137.             _uuid,
138.             "Is_this_house_rented/_owned",
139.             CASE "Highest_Education_Qualification"
140.                 WHEN 'Illiterat/Never attended school' THEN 0
141.                 WHEN 'Primary (class 1 - 5)' THEN 2
142.                 WHEN 'Secondary (class 6 - 10)' THEN 3
143.                 WHEN 'Higher Secondary/ Intermediate (class 11 - 12)'
        THEN 4
144.                 WHEN 'Graduation' THEN 5
145.                 ELSE -1 -- Handle empty or unknown values
146.             END AS EducationRank
147.         FROM "Respondent"
148.     ),
149.     HouseholdMembersEducation AS (
150.         SELECT
151.             _submission__uuid,
152.             CASE "Highest_Education_Qualification"
153.                 WHEN 'Illiterate/Never attended school' THEN 0
154.                 WHEN 'PrePrimary' THEN 1
155.                 WHEN 'Primary (class 1__5)' THEN 2
156.                 WHEN 'Secondary (class 6__10)' THEN 3
157.                 WHEN 'Higher_Secondary/_Intermediate_ (class 11__12)'
        THEN 4
158.                 WHEN 'Graduation' THEN 5
159.                 ELSE -1 -- Handle empty or unknown values
160.             END AS EducationRank
161.         FROM "Household_members"
162.     ),
163.     FamilyHighestEducation AS (
164.         SELECT
165.             R._uuid,
166.             R."Is_this_house_rented/_owned",
167.             MAX(T.EducationRank) AS MaxEducationRank
168.         FROM RespondentEducation AS R
169.         LEFT JOIN (
170.             SELECT _uuid, EducationRank FROM RespondentEducation
171.             UNION ALL
172.             SELECT _submission__uuid AS _uuid, EducationRank FROM
        HouseholdMembersEducation
173.         ) AS T ON R._uuid = T._uuid
174.         GROUP BY R._uuid, R."Is_this_house_rented/_owned"
175.     )
176.     SELECT

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177.         F."Is_this_house_rented/_owned",
178.         SUM(CASE WHEN F.MaxEducationRank = 0 THEN 1 ELSE 0 END) AS
179.         "Illiterate/Never attended school",
180.         SUM(CASE WHEN F.MaxEducationRank = 1 THEN 1 ELSE 0 END) AS
181.         "PrePrimary",
182.         SUM(CASE WHEN F.MaxEducationRank = 2 THEN 1 ELSE 0 END) AS
183.         "Primary (class 1 - 5)",
184.         SUM(CASE WHEN F.MaxEducationRank = 3 THEN 1 ELSE 0 END) AS
185.         "Secondary (class 6 - 10)",
186.         SUM(CASE WHEN F.MaxEducationRank = 4 THEN 1 ELSE 0 END) AS
187.         "Higher Secondary/ Intermediate (class 11 - 12)",
188.         SUM(CASE WHEN F.MaxEducationRank = 5 THEN 1 ELSE 0 END) AS
189.         "Graduation",
190.         SUM(CASE WHEN F.MaxEducationRank = -1 THEN 1 ELSE 0 END) AS
191.         "Unknown/N/A"
192.     FROM FamilyHighestEducation AS F
193.     GROUP BY F."Is_this_house_rented/_owned";```sql
194.     WITH RespondentEducation AS (
195.         SELECT
196.             _uuid,
197.             "Is_this_house_rented/_owned",
198.             CASE "Highest_Education_Qualification"
199.                 WHEN 'Illiterate/Never attended school' THEN 0
200.                 WHEN 'Primary (class 1 - 5)' THEN 2
201.                 WHEN 'Secondary (class 6 - 10)' THEN 3
202.                 WHEN 'Higher Secondary/ Intermediate (class 11 - 12)'
203.             THEN 4
204.                 WHEN 'Graduation' THEN 5
205.                 ELSE -1 -- Handle empty or unknown values
206.             END AS EducationRank
207.         FROM "Respondent"
208.     ),
209.     HouseholdMembersEducation AS (
210.         SELECT
211.             _submission__uuid,
212.             CASE "Highest_Education_Qualification"
213.                 WHEN 'Illiterate/Never attended_school' THEN 0
214.                 WHEN 'PrePrimary' THEN 1
215.                 WHEN 'Primary (class 1_5)' THEN 2
216.                 WHEN 'Secondary (class 6_10)' THEN 3
217.                 WHEN 'Higher_Secondary/_Intermediate_(class_11_12)'
218.             THEN 4
219.                 WHEN 'Graduation' THEN 5
220.                 ELSE -1 -- Handle empty or unknown values
221.             END AS EducationRank
222.         FROM "Household_members"
223.     ),
224.     FamilyHighestEducation AS (
225.         SELECT
226.             R._uuid,
227.             R."Is_this_house_rented/_owned",
228.             MAX(T.EducationRank) AS MaxEducationRank
229.         FROM RespondentEducation AS R
230.         LEFT JOIN (
231.             SELECT _uuid, EducationRank FROM RespondentEducation
232.             UNION ALL
233.             SELECT _submission__uuid AS _uuid, EducationRank FROM
234.             HouseholdMembersEducation
235.         ) AS T ON R._uuid = T._uuid
236.         GROUP BY R._uuid, R."Is_this_house_rented/_owned"
237.     )

```

```

228.     SELECT
229.         F."Is_this_house_rented/_owned",
230.         SUM(CASE WHEN F.MaxEducationRank = 0 THEN 1 ELSE 0 END) AS
231.         "Illiterate/Never attended school",
232.         SUM(CASE WHEN F.MaxEducationRank = 1 THEN 1 ELSE 0 END) AS
233.         "PrePrimary",
234.         SUM(CASE WHEN F.MaxEducationRank = 2 THEN 1 ELSE 0 END) AS
235.         "Primary (class 1 - 5)",
236.         SUM(CASE WHEN F.MaxEducationRank = 3 THEN 1 ELSE 0 END) AS
237.         "Secondary (class 6 - 10)",
238.         SUM(CASE WHEN F.MaxEducationRank = 4 THEN 1 ELSE 0 END) AS
239.         "Higher Secondary/ Intermediate (class 11 - 12)",
240.         SUM(CASE WHEN F.MaxEducationRank = 5 THEN 1 ELSE 0 END) AS
241.         "Graduation",
242.         SUM(CASE WHEN F.MaxEducationRank = -1 THEN 1 ELSE 0 END) AS
243.         "Unknown/N/A"
244.     FROM FamilyHighestEducation AS F
245.     GROUP BY F."Is_this_house_rented/_owned";
246.     ``Thought:The query checker returned the query itself, which means
247.     it's syntactically correct. Now I will execute the query to get the
248.     crosstab results.
249.
250.     Action: sql_db_query
251.     Action Input: WITH RespondentEducation AS (
252.         SELECT
253.             _uuid,
254.             "Is_this_house_rented/_owned",
255.             CASE "Highest_Education_Qualification"
256.                 WHEN 'Illiterat_/Never attended school' THEN 0
257.                 WHEN 'Primary (class 1 - 5)' THEN 2
258.                 WHEN 'Secondary (class 6 - 10)' THEN 3
259.                 WHEN 'Higher Secondary/ Intermediate (class 11 - 12)'
260.                 THEN 4
261.                 WHEN 'Graduation' THEN 5
262.                 ELSE -1 -- Handle empty or unknown values
263.             END AS EducationRank
264.         FROM "Respondent"
265.     ),
266.     HouseholdMembersEducation AS (
267.         SELECT
268.             _submission_uuid,
269.             CASE "Highest_Education_Qualification"
270.                 WHEN 'Illiterate/Never_attended_school' THEN 0
271.                 WHEN 'PrePrimary' THEN 1
272.                 WHEN 'Primary_(class_1_5)' THEN 2
273.                 WHEN 'Secondary_(class_6_10)' THEN 3
274.                 WHEN 'Higher_Secondary/_Intermediate_(class_11_12)'
275.                 THEN 4
276.                 WHEN 'Graduation' THEN 5
277.                 ELSE -1 -- Handle empty or unknown values
278.             END AS EducationRank
279.         FROM "Household_members"
280.     ),
281.     FamilyHighestEducation AS (
282.         SELECT
283.             R._uuid,
284.             R."Is_this_house_rented/_owned",
285.             MAX(T.EducationRank) AS MaxEducationRank
286.         FROM RespondentEducation AS R
287.         LEFT JOIN (
288.             SELECT _uuid, EducationRank FROM RespondentEducation

```

```

278.          UNION ALL
279.          SELECT _submission__uuid AS _uuid, EducationRank FROM
HouseholdMembersEducation
280.          ) AS T ON R._uuid = T._uuid
281.          GROUP BY R._uuid, R."Is_this_house_rented/_owned"
282.      )
283.      SELECT
284.          F."Is_this_house_rented/_owned",
285.          SUM(CASE WHEN F.MaxEducationRank = 0 THEN 1 ELSE 0 END) AS
"Illiterate/Never attended school",
286.          SUM(CASE WHEN F.MaxEducationRank = 1 THEN 1 ELSE 0 END) AS
"PrePrimary",
287.          SUM(CASE WHEN F.MaxEducationRank = 2 THEN 1 ELSE 0 END) AS
"Primary (class 1 - 5)",
288.          SUM(CASE WHEN F.MaxEducationRank = 3 THEN 1 ELSE 0 END) AS
"Secondary (class 6 - 10)",
289.          SUM(CASE WHEN F.MaxEducationRank = 4 THEN 1 ELSE 0 END) AS
"Higher Secondary/ Intermediate (class 11 - 12)",
290.          SUM(CASE WHEN F.MaxEducationRank = 5 THEN 1 ELSE 0 END) AS
"Graduation",
291.          SUM(CASE WHEN F.MaxEducationRank = -1 THEN 1 ELSE 0 END) AS
"Unknown/N/A"
292.      FROM FamilyHighestEducation AS F
293.      GROUP BY F."Is_this_house_rented/_owned";[('Own', 0, 0, 0, 2, 2,
6, 0), ('Rent', 0, 0, 1, 1, 0, 1, 1), ('Squatting', 1, 0, 0, 5, 1, 7,
0), ('Temporarily staying as Guests in Relative/Friends place', 0, 0,
0, 1, 0, 0, 0)]Thought:I have both parts of the answer.
294.      Part 1: Count of respondents by housing tenure.
295.          - Own: 10
296.          - Rent: 4
297.          - Squatting: 14
298.          - Temporarily staying as Guests in Relative/Friends place: 1
299.
300.      Part 2: Crosstab of housing tenure by highest educational
qualification in the respondents' families.
301.          - **Own**:
302.              - Illiterate/Never attended school: 0
303.              - PrePrimary: 0
304.              - Primary (class 1 - 5): 0
305.              - Secondary (class 6 - 10): 2
306.              - Higher Secondary/ Intermediate (class 11 - 12): 2
307.              - Graduation: 6
308.              - Unknown/N/A: 0
309.          - **Rent**:
310.              - Illiterate/Never attended school: 0
311.              - PrePrimary: 0
312.              - Primary (class 1 - 5): 1
313.              - Secondary (class 6 - 10): 1
314.              - Higher Secondary/ Intermediate (class 11 - 12): 0
315.              - Graduation: 1
316.              - Unknown/N/A: 1
317.          - **Squatting**:
318.              - Illiterate/Never attended school: 1
319.              - PrePrimary: 0
320.              - Primary (class 1 - 5): 0
321.              - Secondary (class 6 - 10): 5
322.              - Higher Secondary/ Intermediate (class 11 - 12): 1
323.              - Graduation: 7
324.              - Unknown/N/A: 0
325.          - **Temporarily staying as Guests in Relative/Friends place**:
326.              - Illiterate/Never attended school: 0

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327.      - PrePrimary: 0
328.      - Primary (class 1 - 5): 0
329.      - Secondary (class 6 - 10): 1
330.      - Higher Secondary/ Intermediate (class 11 - 12): 0
331.      - Graduation: 0
332.      - Unknown/N/A: 0
333.
334.      I will now format the final answer.Final Answer:
335.      **Respondent Housing Tenure Counts:**
336.      - Own: 10 respondents
337.      - Rent: 4 respondents
338.      - Squatting: 14 respondents
339.      - Temporarily staying as Guests in Relative_/Friends place: 1
        respondent
340.
341.      **Crosstab of Housing Tenure by Highest Family Educational
        Qualification:**
342.      - **Own:**
343.          - Secondary (class 6 - 10): 2 families
344.          - Higher Secondary/ Intermediate (class 11 - 12): 2 families
345.          - Graduation: 6 families
346.      - **Rent:**
347.          - Primary (class 1 - 5): 1 family
348.          - Secondary (class 6 - 10): 1 family
349.          - Graduation: 1 family
350.          - Unknown/N/A: 1 family
351.      - **Squatting:**
352.          - Illiterate/Never attended school: 1 family
353.          - Secondary (class 6 - 10): 5 families
354.          - Higher Secondary/ Intermediate (class 11 - 12): 1 family
355.          - Graduation: 7 families
356.      - **Temporarily staying as Guests in Relative_/Friends place:**
357.          - Secondary (class 6 - 10): 1 family
358.
359.      > Finished chain.
360.
361.      =====
362.      🇳🇵 FINAL ANSWER
363.      =====
364.      **Respondent Housing Tenure Counts:**
365.      - Own: 10 respondents
366.      - Rent: 4 respondents
367.      - Squatting: 14 respondents
368.      - Temporarily staying as Guests in Relative_/Friends place: 1
        respondent
369.
370.      **Crosstab of Housing Tenure by Highest Family Educational
        Qualification:**
371.      - **Own:**
372.          - Secondary (class 6 - 10): 2 families
373.          - Higher Secondary/ Intermediate (class 11 - 12): 2 families
374.          - Graduation: 6 families
375.      - **Rent:**
376.          - Primary (class 1 - 5): 1 family
377.          - Secondary (class 6 - 10): 1 family
378.          - Graduation: 1 family
379.          - Unknown/N/A: 1 family
380.      - **Squatting:**
381.          - Illiterate/Never attended school: 1 family
382.          - Secondary (class 6 - 10): 5 families
383.          - Higher Secondary/ Intermediate (class 11 - 12): 1 family

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384. - Graduation: 7 families
385. - **Temporarily staying as Guests in Relative_/Friends place:**
386. - Secondary (class 6 - 10): 1 family
387.
388. Query (or 'exit'): exit
389.
390. Session ended. Goodbye!

3. What is the composition (%age) of all primary and secondary occupations of the respondents and their families?

WITH RespondentOccupations AS (

SELECT 'Engineers' AS Occupation, CAST("Primary_Occupation_/Engineers" AS REAL) AS Count FROM "Respondent" WHERE "Primary_Occupation_/Engineers" = '1'

UNION ALL SELECT 'Store Managers' AS Occupation, CAST("Primary_Occupation_/Store_Managers" AS REAL) AS Count FROM "Respondent" WHERE "Primary_Occupation_/Store_Managers" = '1'

UNION ALL SELECT 'Teachers' AS Occupation, CAST("Primary_Occupation_/Teachers" AS REAL) AS Count FROM "Respondent" WHERE "Primary_Occupation_/Teachers" = '1'

UNION ALL SELECT 'GHMC Workers' AS Occupation, CAST("Primary_Occupation_/GHMC_Workers" AS REAL) AS Count FROM "Respondent" WHERE "Primary_Occupation_/GHMC_Workers" = '1'

UNION ALL SELECT 'Construction Workers/Maestri' AS Occupation, CAST("Primary_Occupation_/Construction_worker_/Maestri" AS REAL) AS Count FROM "Respondent" WHERE "Primary_Occupation_/Construction_worker_/Maestri" = '1'

UNION ALL SELECT 'Painter' AS Occupation, CAST("Primary_Occupation_/Painter" AS REAL) AS Count FROM "Respondent" WHERE "Primary_Occupation_/Painter" = '1'

UNION ALL SELECT 'Security Guards' AS Occupation, CAST("Primary_Occupation_/Security_Guards" AS REAL) AS Count FROM "Respondent" WHERE "Primary_Occupation_/Security_Guards" = '1'

UNION ALL SELECT 'Carpenter' AS Occupation, CAST("Primary_Occupation_/Carpenter" AS REAL) AS Count FROM "Respondent" WHERE "Primary_Occupation_/Carpenter" = '1'

UNION ALL SELECT 'Electrician' AS Occupation, CAST("Primary_Occupation_/Electrician" AS REAL) AS Count FROM "Respondent" WHERE "Primary_Occupation_/Electrician" = '1'

UNION ALL SELECT 'Shop Keeper' AS Occupation, CAST("Primary_Occupation_/Shop_Keeper" AS REAL) AS Count FROM "Respondent" WHERE "Primary_Occupation_/Shop_Keeper" = '1'

UNION ALL SELECT 'Housekeeping' AS Occupation, CAST("Primary_Occupation_/Housekeeping_(working_in_malls__corporate_offices)" AS REAL) AS Count FROM "Respondent" WHERE "Primary_Occupation_/Housekeeping_(working_in_malls__corporate_offices)" = '1'

UNION ALL SELECT 'Maids/Cooks' AS Occupation, CAST("Primary_Occupation_/Maid_/Cooks" AS REAL) AS Count FROM "Respondent" WHERE "Primary_Occupation_/Maid_/Cooks" = '1'

UNION ALL SELECT 'Auto/Cab/Truck/Car Drivers' AS Occupation, CAST("Primary_Occupation_/Aut_/Ca_/Truc_/Car_Drivers" AS REAL) AS Count FROM "Respondent" WHERE "Primary_Occupation_/Aut_/Ca_/Truc_/Car_Drivers" = '1'

UNION ALL SELECT 'Street Vendors' AS Occupation, CAST("Primary_Occupation_/Street_Vendors" AS REAL) AS Count FROM "Respondent" WHERE "Primary_Occupation_/Street_Vendors" = '1'

UNION ALL SELECT 'Bouncers' AS Occupation, CAST("Primary_Occupation_/Bouncers" AS REAL) AS Count FROM "Respondent" WHERE "Primary_Occupation_/Bouncers" = '1'

UNION ALL SELECT 'Housewife' AS Occupation, CAST("Primary_Occupation_/Housewife" AS REAL) AS Count FROM "Respondent" WHERE "Primary_Occupation_/Housewife" = '1'

UNION ALL SELECT 'Pensioner' AS Occupation, CAST("Primary_Occupation_/Pensioner" AS REAL) AS Count FROM "Respondent" WHERE "Primary_Occupation_/Pensioner" = '1'

UNION ALL SELECT 'Garbage Segregation' AS Occupation, CAST("Primary_Occupation_/Garbage_Segregation" AS REAL) AS Count FROM "Respondent" WHERE "Primary_Occupation_/Garbage_Segregation" = '1'

UNION ALL SELECT 'Garlic Peeling' AS Occupation, CAST("Primary_Occupation_/Garlic_peeling" AS REAL) AS Count FROM "Respondent" WHERE "Primary_Occupation_/Garlic_peeling" = '1'

UNION ALL SELECT 'Student' AS Occupation, CAST("Primary_Occupation_/Student" AS REAL) AS Count FROM "Respondent" WHERE "Primary_Occupation_/Student" = '1'

UNION ALL SELECT 'Others' AS Occupation, CAST("Primary_Occupation_/Others" AS REAL) AS Count FROM "Respondent" WHERE "Primary_Occupation_/Others" = '1'

--Exclude 'Not engaged in anything' from the list of occupations

```
UNION ALL SELECT 'Not engaged in anything' AS Occupation, CAST("Primary_Occupation_/Not_engaged_in_anything" AS REAL) AS Count
FROM "Respondent" WHERE "Primary_Occupation_/Not_engaged_in_anything" = '1'
```

```
UNION ALL SELECT 'Engineers' AS Occupation, CAST("Secondary_occupation_/Engineers" AS REAL) AS Count FROM "Respondent" WHERE
"Secondary_occupation_/Engineers" = '1'
```

```
UNION ALL SELECT 'Store Managers' AS Occupation, CAST("Secondary_occupation_/Store_Managers" AS REAL) AS Count FROM
"Respondent" WHERE "Secondary_occupation_/Store_Managers" = '1'
```

```
UNION ALL SELECT 'Teachers' AS Occupation, CAST("Secondary_occupation_/Teachers" AS REAL) AS Count FROM "Respondent" WHERE
"Secondary_occupation_/Teachers" = '1'
```

```
UNION ALL SELECT 'GHMC Workers' AS Occupation, CAST("Secondary_occupation_/GHMC_Workers" AS REAL) AS Count FROM "Respondent"
WHERE "Secondary_occupation_/GHMC_Workers" = '1'
```

```
UNION ALL SELECT 'Construction Workers/Maestri' AS Occupation, CAST("Secondary_occupation_/Construction_worker_/Maestri" AS REAL)
AS Count FROM "Respondent" WHERE "Secondary_occupation_/Construction_worker_/Maestri" = '1'
```

```
UNION ALL SELECT 'Painter' AS Occupation, CAST("Secondary_occupation_/Painter" AS REAL) AS Count FROM "Respondent" WHERE
"Secondary_occupation_/Painter" = '1'
```

```
UNION ALL SELECT 'Security Guards' AS Occupation, CAST("Secondary_occupation_/Security_Guards" AS REAL) AS Count FROM
"Respondent" WHERE "Secondary_occupation_/Security_Guards" = '1'
```

```
UNION ALL SELECT 'Carpenter' AS Occupation, CAST("Secondary_occupation_/Carpenter" AS REAL) AS Count FROM "Respondent" WHERE
"Secondary_occupation_/Carpenter" = '1'
```

```
UNION ALL SELECT 'Electrician' AS Occupation, CAST("Secondary_occupation_/Electrician" AS REAL) AS Count FROM "Respondent" WHERE
"Secondary_occupation_/Electrician" = '1'
```

```
UNION ALL SELECT 'Shop Keeper' AS Occupation, CAST("Secondary_occupation_/Shop_Keeper" AS REAL) AS Count FROM "Respondent"
WHERE "Secondary_occupation_/Shop_Keeper" = '1'
```

```
UNION ALL SELECT 'Housekeeping' AS Occupation, CAST("Secondary_occupation_/Housekeeping_(working_in_malls__corporate_offices)" AS
REAL) AS Count FROM "Respondent" WHERE "Secondary_occupation_/Housekeeping_(working_in_malls__corporate_offices)" = '1'
```

```
UNION ALL SELECT 'Maids/Cooks' AS Occupation, CAST("Secondary_occupation_/Maid_/Cooks" AS REAL) AS Count FROM "Respondent"
WHERE "Secondary_occupation_/Maid_/Cooks" = '1'
```

```
UNION ALL SELECT 'Auto/Cab/Truck/Car Drivers' AS Occupation, CAST("Secondary_occupation_/Aut_/Ca_/Truc_/Car_Drivers" AS REAL) AS
Count FROM "Respondent" WHERE "Secondary_occupation_/Aut_/Ca_/Truc_/Car_Drivers" = '1'
```

```
UNION ALL SELECT 'Street Vendors' AS Occupation, CAST("Secondary_occupation_/Street_Vendors" AS REAL) AS Count FROM "Respondent"
WHERE "Secondary_occupation_/Street_Vendors" = '1'
```

```
UNION ALL SELECT 'Bouncers' AS Occupation, CAST("Secondary_occupation_/Bouncers" AS REAL) AS Count FROM "Respondent" WHERE
"Secondary_occupation_/Bouncers" = '1'
```

```
UNION ALL SELECT 'Housewife' AS Occupation, CAST("Secondary_occupation_/Housewife" AS REAL) AS Count FROM "Respondent" WHERE
"Secondary_occupation_/Housewife" = '1'
```

```
UNION ALL SELECT 'Pensioner' AS Occupation, CAST("Secondary_occupation_/Pensioner" AS REAL) AS Count FROM "Respondent" WHERE
"Secondary_occupation_/Pensioner" = '1'
```

```
UNION ALL SELECT 'Garbage Segregation' AS Occupation, CAST("Secondary_occupation_/Garbage_Segregation" AS REAL) AS Count FROM
"Respondent" WHERE "Secondary_occupation_/Garbage_Segregation" = '1'
```

```
UNION ALL SELECT 'Student' AS Occupation, CAST("Secondary_occupation_/Student" AS REAL) AS Count FROM "Respondent" WHERE
"Secondary_occupation_/Student" = '1'
```

```
UNION ALL SELECT 'Others' AS Occupation, CAST("Secondary_Occupation_/Others" AS REAL) AS Count FROM "Respondent" WHERE
"Secondary_Occupation_/Others" = '1'
```

-- Exclude 'Not engaged in anything' and 'N/A' from the list of occupations

```
UNION ALL SELECT 'Not engaged in anything' AS Occupation, CAST("Secondary_Occupation_/Not_engaged_in_anything" AS REAL) AS Count
FROM "Respondent" WHERE "Secondary_Occupation_/Not_engaged_in_anything" = '1'
```

```
UNION ALL SELECT 'N/A' AS Occupation, CAST("Secondary_Occupation_/NA" AS REAL) AS Count FROM "Respondent" WHERE
"Secondary_Occupation_/NA" = '1'
```

UNION ALL SELECT 'Ginger/Garlic Peeling' AS Occupation, CAST("Secondary_occupation_/Ginge_/garlic_peeling" AS REAL) AS Count FROM "Respondent" WHERE "Secondary_occupation_/Ginge_/garlic_peeling" = '1'

),

HouseholdOccupations AS (

SELECT 'Engineers' AS Occupation, CAST("Primary_Occupation/Engineers" AS REAL) AS Count FROM "Household_members" WHERE "Primary_Occupation/Engineers" = '1'

UNION ALL SELECT 'Store Managers' AS Occupation, CAST("Primary_Occupation/Store_Managers" AS REAL) AS Count FROM "Household_members" WHERE "Primary_Occupation/Store_Managers" = '1'

UNION ALL SELECT 'Teachers' AS Occupation, CAST("Primary_Occupation/Teachers" AS REAL) AS Count FROM "Household_members" WHERE "Primary_Occupation/Teachers" = '1'

UNION ALL SELECT 'GHMC Workers' AS Occupation, CAST("Primary_Occupation/GHMC_Workers" AS REAL) AS Count FROM "Household_members" WHERE "Primary_Occupation/GHMC_Workers" = '1'

UNION ALL SELECT 'Construction Workers/Maestri' AS Occupation, CAST("Primary_Occupation/Construction_workers/Maestri" AS REAL) AS Count FROM "Household_members" WHERE "Primary_Occupation/Construction_workers/Maestri" = '1'

UNION ALL SELECT 'Painter' AS Occupation, CAST("Primary_Occupation/Painter" AS REAL) AS Count FROM "Household_members" WHERE "Primary_Occupation/Painter" = '1'

UNION ALL SELECT 'Security Guards' AS Occupation, CAST("Primary_Occupation/Security_Guards" AS REAL) AS Count FROM "Household_members" WHERE "Primary_Occupation/Security_Guards" = '1'

UNION ALL SELECT 'Carpenter' AS Occupation, CAST("Primary_Occupation/Carpenter" AS REAL) AS Count FROM "Household_members" WHERE "Primary_Occupation/Carpenter" = '1'

UNION ALL SELECT 'Electrician' AS Occupation, CAST("Primary_Occupation/Electrician" AS REAL) AS Count FROM "Household_members" WHERE "Primary_Occupation/Electrician" = '1'

UNION ALL SELECT 'Shop Keeper' AS Occupation, CAST("Primary_Occupation/Shop_Keeper" AS REAL) AS Count FROM "Household_members" WHERE "Primary_Occupation/Shop_Keeper" = '1'

UNION ALL SELECT 'Housekeeping' AS Occupation, CAST("Primary_Occupation/Housekeeping_(working_in_malls_corporate_offices)" AS REAL) AS Count FROM "Household_members" WHERE "Primary_Occupation/Housekeeping_(working_in_malls_corporate_offices)" = '1'

UNION ALL SELECT 'Maids/Cooks' AS Occupation, CAST("Primary_Occupation/Maids/Cooks" AS REAL) AS Count FROM "Household_members" WHERE "Primary_Occupation/Maids/Cooks" = '1'

UNION ALL SELECT 'Auto/Cab/Truck/Car Drivers' AS Occupation, CAST("Primary_Occupation/Auto/Cab/Truck/Car_Drivers" AS REAL) AS Count FROM "Household_members" WHERE "Primary_Occupation/Auto/Cab/Truck/Car_Drivers" = '1'

UNION ALL SELECT 'Street Vendors' AS Occupation, CAST("Primary_Occupation/Street_Vendors" AS REAL) AS Count FROM "Household_members" WHERE "Primary_Occupation/Street_Vendors" = '1'

UNION ALL SELECT 'Bouncers' AS Occupation, CAST("Primary_Occupation/Bouncers" AS REAL) AS Count FROM "Household_members" WHERE "Primary_Occupation/Bouncers" = '1'

UNION ALL SELECT 'Housewife' AS Occupation, CAST("Primary_Occupation/Housewife" AS REAL) AS Count FROM "Household_members" WHERE "Primary_Occupation/Housewife" = '1'

UNION ALL SELECT 'Pensioner' AS Occupation, CAST("Primary_Occupation/Pensioner" AS REAL) AS Count FROM "Household_members" WHERE "Primary_Occupation/Pensioner" = '1'

UNION ALL SELECT 'Garbage Segregation' AS Occupation, CAST("Primary_Occupation/Garbage_Segregation" AS REAL) AS Count FROM "Household_members" WHERE "Primary_Occupation/Garbage_Segregation" = '1'

UNION ALL SELECT 'Garlic Peeling' AS Occupation, CAST("Primary_Occupation/Garlic_peeling" AS REAL) AS Count FROM "Household_members" WHERE "Primary_Occupation/Garlic_peeling" = '1'

UNION ALL SELECT 'Student' AS Occupation, CAST("Primary_Occupation/Student" AS REAL) AS Count FROM "Household_members" WHERE "Primary_Occupation/Student" = '1'

UNION ALL SELECT 'Others' AS Occupation, CAST("Primary_Occupation/Others" AS REAL) AS Count FROM "Household_members" WHERE "Primary_Occupation/Others" = '1'

-- Exclude 'Not engaged in anything' and 'I'm not sure'

UNION ALL SELECT 'Not engaged in anything' AS Occupation, CAST("Primary_Occupation/Not_engaged_in_anything" AS REAL) AS Count FROM "Household_members" WHERE "Primary_Occupation/Not_engaged_in_anything" = '1'

-- UNION ALL SELECT 'I'm not sure' AS Occupation, CAST("Primary_Occupation/I'm_not_sure" AS REAL) AS Count FROM "Household_members" WHERE "Primary_Occupation/I'm_not_sure" = '1'

UNION ALL SELECT 'Engineers' AS Occupation, CAST("Subsidiary_Occupations_of_Household_Members/Engineers" AS REAL) AS Count FROM "Household_members" WHERE "Subsidiary_Occupations_of_Household_Members/Engineers" = '1'

UNION ALL SELECT 'Store Managers' AS Occupation, CAST("Subsidiary_Occupations_of_Household_Members/Store_Managers" AS REAL) AS Count FROM "Household_members" WHERE "Subsidiary_Occupations_of_Household_Members/Store_Managers" = '1'

UNION ALL SELECT 'Teachers' AS Occupation, CAST("Subsidiary_Occupations_of_Household_Members/Teachers" AS REAL) AS Count FROM "Household_members" WHERE "Subsidiary_Occupations_of_Household_Members/Teachers" = '1'

UNION ALL SELECT 'GHMC Workers' AS Occupation, CAST("Subsidiary_Occupations_of_Household_Members/GHMC_Workers" AS REAL) AS Count FROM "Household_members" WHERE "Subsidiary_Occupations_of_Household_Members/GHMC_Workers" = '1'

UNION ALL SELECT 'Construction Workers/Maestri' AS Occupation, CAST("Subsidiary_Occupations_of_Household_Members/Construction_workers/Maestri" AS REAL) AS Count FROM "Household_members" WHERE "Subsidiary_Occupations_of_Household_Members/Construction_workers/Maestri" = '1'

UNION ALL SELECT 'Painter' AS Occupation, CAST("Subsidiary_Occupations_of_Household_Members/Painter" AS REAL) AS Count FROM "Household_members" WHERE "Subsidiary_Occupations_of_Household_Members/Painter" = '1'

UNION ALL SELECT 'Security Guards' AS Occupation, CAST("Subsidiary_Occupations_of_Household_Members/Security_Guards" AS REAL) AS Count FROM "Household_members" WHERE "Subsidiary_Occupations_of_Household_Members/Security_Guards" = '1'

UNION ALL SELECT 'Carpenter' AS Occupation, CAST("Subsidiary_Occupations_of_Household_Members/Carpenter" AS REAL) AS Count FROM "Household_members" WHERE "Subsidiary_Occupations_of_Household_Members/Carpenter" = '1'

UNION ALL SELECT 'Electrician' AS Occupation, CAST("Subsidiary_Occupations_of_Household_Members/Electrician" AS REAL) AS Count FROM "Household_members" WHERE "Subsidiary_Occupations_of_Household_Members/Electrician" = '1'

UNION ALL SELECT 'Shop Keeper' AS Occupation, CAST("Subsidiary_Occupations_of_Household_Members/Shop_Keeper" AS REAL) AS Count FROM "Household_members" WHERE "Subsidiary_Occupations_of_Household_Members/Shop_Keeper" = '1'

UNION ALL SELECT 'Housekeeping' AS Occupation, CAST("Subsidiary_Occupations_of_Household_Members/Housekeeping_(working_in_malls_corporate_offices)" AS REAL) AS Count FROM "Household_members" WHERE "Subsidiary_Occupations_of_Household_Members/Housekeeping_(working_in_malls_corporate_offices)" = '1'

UNION ALL SELECT 'Maids/Cooks' AS Occupation, CAST("Subsidiary_Occupations_of_Household_Members/Maids/Cooks" AS REAL) AS Count FROM "Household_members" WHERE "Subsidiary_Occupations_of_Household_Members/Maids/Cooks" = '1'

UNION ALL SELECT 'Auto/Cab/Truck/Car Drivers' AS Occupation, CAST("Subsidiary_Occupations_of_Household_Members/Auto/Cab/Truck/Car_Drivers" AS REAL) AS Count FROM "Household_members" WHERE "Subsidiary_Occupations_of_Household_Members/Auto/Cab/Truck/Car_Drivers" = '1'

UNION ALL SELECT 'Street Vendors' AS Occupation, CAST("Subsidiary_Occupations_of_Household_Members/Street_Vendors" AS REAL) AS Count FROM "Household_members" WHERE "Subsidiary_Occupations_of_Household_Members/Street_Vendors" = '1'

UNION ALL SELECT 'Bouncers' AS Occupation, CAST("Subsidiary_Occupations_of_Household_Members/Bouncers" AS REAL) AS Count FROM "Household_members" WHERE "Subsidiary_Occupations_of_Household_Members/Bouncers" = '1'

UNION ALL SELECT 'Housewife' AS Occupation, CAST("Subsidiary_Occupations_of_Household_Members/Housewife" AS REAL) AS Count FROM "Household_members" WHERE "Subsidiary_Occupations_of_Household_Members/Housewife" = '1'

UNION ALL SELECT 'Pensioner' AS Occupation, CAST("Subsidiary_Occupations_of_Household_Members/Pensioner" AS REAL) AS Count FROM "Household_members" WHERE "Subsidiary_Occupations_of_Household_Members/Pensioner" = '1'

UNION ALL SELECT 'Garbage Segregation' AS Occupation, CAST("Subsidiary_Occupations_of_Household_Members/Garbage_Segregation" AS REAL) AS Count FROM "Household_members" WHERE "Subsidiary_Occupations_of_Household_Members/Garbage_Segregation" = '1'

UNION ALL SELECT 'Student' AS Occupation, CAST("Subsidiary_Occupations_of_Household_Members/Student" AS REAL) AS Count FROM "Household_members" WHERE "Subsidiary_Occupations_of_Household_Members/Student" = '1'

UNION ALL SELECT 'Others' AS Occupation, CAST("Subsidiary_Occupations_of_Household_Members/Others" AS REAL) AS Count FROM "Household_members" WHERE "Subsidiary_Occupations_of_Household_Members/Others" = '1'

-- Exclude 'Not engaged in anything', 'N/A', and 'I'm not sure'

UNION ALL SELECT 'Not engaged in anything' AS Occupation, CAST("Subsidiary_Occupations_of_Household_Members/Not_engaged_in_anything" AS REAL) AS Count FROM "Household_members" WHERE "Subsidiary_Occupations_of_Household_Members/Not_engaged_in_anything" = '1'

```

UNION ALL SELECT 'N/A' AS Occupation, CAST("Subsidiary_Occupations_of_Household_Members/N/A" AS REAL) AS Count FROM
"Household_members" WHERE "Subsidiary_Occupations_of_Household_Members/N/A" = '1'

-- UNION ALL SELECT 'I'm not sure' AS Occupation, CAST("Subsidiary_Occupations_of_Household_Members/I'm_not_sure" AS REAL) AS
Count FROM "Household_members" WHERE "Subsidiary_Occupations_of_Household_Members/I'm_not_sure" = '1'

UNION ALL SELECT 'Ginger/Garlic Peeling' AS Occupation, CAST("Subsidiary_Occupations_of_Household_Members/Ginger/garlic_peeling" AS
REAL) AS Count FROM "Household_members" WHERE "Subsidiary_Occupations_of_Household_Members/Ginger/garlic_peeling" = '1'

),

AllOccupations AS (

SELECT Occupation, Count FROM RespondentOccupations

UNION ALL

SELECT Occupation, Count FROM HouseholdOccupations

)

SELECT

Occupation,

SUM(Count) AS TotalCount,

ROUND(SUM(Count) * 100.0 / (SELECT SUM(Count) FROM AllOccupations), 2) AS Percentage

FROM AllOccupations

GROUP BY Occupation

ORDER BY TotalCount DESC;

```

4.2. also include occupation from if others then specify column also.

```

WITH AllOccupations AS (

-- Respondent Primary Occupations

SELECT

CASE

WHEN T1."Primary_Occupation" = 'Others' AND T1."Primary_Occupation_/If_Other__specify" IS NOT NULL AND
T1."Primary_Occupation_/If_Other__specify" != " THEN T1."Primary_Occupation_/If_Other__specify"

WHEN T1."Primary_Occupation" IS NOT NULL AND T1."Primary_Occupation" != " THEN T1."Primary_Occupation"

ELSE NULL

END AS occupation

FROM "Respondent" AS T1

UNION ALL

-- Respondent Secondary Occupations

SELECT

CASE

WHEN T1."Secondary_Occupation" = 'Others' AND T1."Secondary_Occupation_/If_Other__specify" IS NOT NULL AND
T1."Secondary_Occupation_/If_Other__specify" != " THEN T1."Secondary_Occupation_/If_Other__specify"

```

```

WHEN T1."Secondary_Occupation" IS NOT NULL AND T1."Secondary_Occupation" != '' AND T1."Secondary_Occupation" != '_/A' THEN
T1."Secondary_Occupation"

ELSE NULL

END AS occupation

FROM "Respondent" AS T1

UNION ALL

-- Household_members Primary Occupations

SELECT

CASE

WHEN T2."Primary_Occupation" = 'Others' AND T2."Primary_Occupation/If_Other_specify," IS NOT NULL AND
T2."Primary_Occupation/If_Other_specify," != '' THEN T2."Primary_Occupation/If_Other_specify,"

WHEN T2."Primary_Occupation" IS NOT NULL AND T2."Primary_Occupation" != '' THEN T2."Primary_Occupation"

ELSE NULL

END AS occupation

FROM "Household_members" AS T2

UNION ALL

-- Household_members Secondary Occupations

SELECT

CASE

WHEN T2."Subsidiary_Occupations_of_Household_Members" = 'Others' AND
T2."Subsidiary_Occupations_of_Household_Members/If_Other_specify," IS NOT NULL AND
T2."Subsidiary_Occupations_of_Household_Members/If_Other_specify," != '' THEN
T2."Subsidiary_Occupations_of_Household_Members/If_Other_specify,"

WHEN T2."Subsidiary_Occupations_of_Household_Members" IS NOT NULL AND T2."Subsidiary_Occupations_of_Household_Members" != ''
AND T2."Subsidiary_Occupations_of_Household_Members" != 'N/A' THEN T2."Subsidiary_Occupations_of_Household_Members"

ELSE NULL

END AS occupation

FROM "Household_members" AS T2

)

SELECT

occupation,

COUNT(*) AS count,

ROUND(CAST(COUNT(*) AS REAL) * 100 / (SELECT COUNT(*) FROM AllOccupations WHERE occupation IS NOT NULL AND occupation != ''
), 2) AS percentage

FROM AllOccupations

WHERE occupation IS NOT NULL AND occupation != ''

GROUP BY occupation

ORDER BY count DESC;

```

5. What is the monthly income of the respondents' household? Categorize by religion and caste.

--- SQL Agent Interactive Mode ---

Type your database question and press Enter. Type 'exit' or 'quit' to end the session.

Query (or 'exit'): What is the monthly income of the respondents' household? Categorize by religion and caste.

--- Processing Query: What is the monthly income of the respondents' household? Categorize by religion and caste. --
-

> Entering new SQL Agent Executor chain...

Action: *sql_db_query_checker*

Action Input: *SELECT "Religion", "Category_of_Caste",
AVG(CAST(NULLIF("What_is_the_monthly_income_of_your_household_on_an_average_", "") AS REAL)) AS
"Average_Monthly_Income" FROM "Respondent" GROUP BY "Religion", "Category_of_Caste" ORDER BY
"Religion", "Category_of_Caste";SELECT "Religion", "Category_of_Caste",
AVG(CAST(NULLIF("What_is_the_monthly_income_of_your_household_on_an_average_", "") AS REAL)) AS
"Average_Monthly_Income" FROM "Respondent" GROUP BY "Religion", "Category_of_Caste" ORDER BY
"Religion", "Category_of_Caste";Action: sql_db_query*

Action Input: *SELECT "Religion", "Category_of_Caste",
AVG(CAST(NULLIF("What_is_the_monthly_income_of_your_household_on_an_average_", "") AS REAL)) AS
"Average_Monthly_Income" FROM "Respondent" GROUP BY "Religion", "Category_of_Caste" ORDER BY
"Religion", "Category_of_Caste";[(, 'OBC', 20000.0), (, 'SC', None), ('Christian', 'OBC', 20000.0), ('Christian', 'SC',
15000.0), ('Hindu', , 26250.0), ('Hindu', 'General', 10000.0), ('Hindu', 'OBC', 26000.0), ('Hindu', 'SC',
11666.666666666666), ('Hindu', 'ST', 18333.333333333332), ('Muslim', , 5500.0), ('Muslim', 'General', 24000.0),
('Prefer Not to Say', 'SC', 10000.0), ('Prefer Not to Say', 'ST', 11000.0)]I have successfully retrieved the average
monthly household income categorized by religion and caste. The results show various average incomes
across different religious and caste groups, with some categories having no data (None).*

Here's a summary of the average monthly incomes:

- For no specified religion: OBC has an average of 20000.0.

- For Christian: OBC has an average of 20000.0, and SC has an average of 15000.0.

- *For Hindu: no specified caste has an average of 26250.0, General has 10000.0, OBC has 26000.0, SC has approximately 11666.67, and ST has approximately 18333.33.*

- *For Muslim: no specified caste has an average of 5500.0, and General has 24000.0.*

- *For 'Prefer Not to Say' religion: SC has an average of 10000.0, and ST has 11000.0.*

Final Answer: The average monthly household income of respondents, categorized by religion and caste, is as follows:

- ****No Religion Specified:** OBC: 20000.0**

- ****Christian:** OBC: 20000.0, SC: 15000.0**

- ****Hindu:** No Caste Specified: 26250.0, General: 10000.0, OBC: 26000.0, SC: 11666.67, ST: 18333.33**

- ****Muslim:** No Caste Specified: 5500.0, General: 24000.0**

- ****Prefer Not to Say:** SC: 10000.0, ST: 11000.0**

> Finished chain.

=====
=====

The average monthly household income of respondents, categorized by religion and caste, is as follows:

- ****No Religion Specified:** OBC: 20000.0**

- ****Christian:** OBC: 20000.0, SC: 15000.0**

- ****Hindu:** No Caste Specified: 26250.0, General: 10000.0, OBC: 26000.0, SC: 11666.67, ST: 18333.33**

- ****Muslim:** No Caste Specified: 5500.0, General: 24000.0**

- ****Prefer Not to Say:** SC: 10000.0, ST: 11000.0**

Query (or 'exit'): exit

Session ended. Goodbye!

	Religion	Category_of_Caste	Number of Respondents	Average Monthly Income
1		OBC	1	0
2		SC	1	0
3	Christian	OBC	2	0
4	Christian	SC	1	0
5	Hindu		4	0
6	Hindu	General	1	0
7	Hindu	OBC	4	0
8	Hindu	SC	4	0
9	Hindu	ST	6	0
10	Muslim		2	0
11	Muslim	General	1	0

6. What is the composition (%age) of all chronic diseases of the respondents and their families?

--- SQL Agent Interactive Mode ---

Type your database question and press Enter. Type 'exit' or 'quit' to end the session.

Query (or 'exit'): What is the composition (%age) of all chronic diseases of the respondents and their families? take illnesses from household members and also from if others than specify column.

--- Processing Query: What is the composition (%age) of all chronic diseases of the respondents and their families? take illnesses from household members and also from if others than specify column. ---

> Entering new SQL Agent Executor chain...

Thought: I need to calculate the percentage composition of all chronic diseases for both respondents and household members. This requires combining data from both tables, including specific disease columns and parsing the 'Other' specified illnesses.

My plan is to:

1. Create a series of `UNION ALL` statements for each specific chronic illness from the "Respondent" table, counting occurrences where the corresponding column is '1'.

2. Include `CASE` statements within the "Respondent" section to extract specific 'Other' illnesses from the "Any_Chronic_Illnes_If_Other_specify" column when "Any_Chronic_Illnes_Others" is '1'.

3. Repeat steps 1 and 2 for the "Household_members" table, using its specific column names.
4. Combine all these `UNION ALL` results into a Common Table Expression (CTE) named `AllDiseases`.
5. Group the results by disease name, sum the counts, and calculate the percentage of each disease relative to the total number of reported illnesses.
6. Filter out any null disease entries and diseases with a total count of 0.
7. Order the results by total count in descending order.

Action: `sql_db_query_checker`

Action Input: ``sqlite

WITH AllDiseases AS (

-- Diseases from Respondent table

SELECT 'Tuberculosis' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Tuberculosis" = '1'

UNION ALL

SELECT 'Arthritis/Back Pain' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Arthriti_/Back_Pain" = '1'

UNION ALL

SELECT 'Diabetes' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Diabetes" = '1'

UNION ALL

SELECT 'Paralysis' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Paralysis" = '1'

UNION ALL

SELECT 'High BP' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/High_BP" = '1'

UNION ALL

SELECT 'Heart Disease and stroke' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Heart_Disease_and_stroke" = '1'

UNION ALL

SELECT 'Cancer' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Cancer" = '1'

UNION ALL

SELECT 'Alzheimer's Disease' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Alzheimer's_Disease" = '1'

UNION ALL

SELECT 'Kidney Diseases' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Kidney_Diseases" = '1'

UNION ALL

SELECT 'Liver Cirrhosis' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Liver_Cirrhosis" = '1'

UNION ALL

SELECT 'Peptic Ulcer (stomach)' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Peptic_Ulcer(stomach)" = '1'

UNION ALL

SELECT 'Alcohol related health issues' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Alcohol_related_health_issues" = '1'

UNION ALL

SELECT 'Smoking related health issues' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Smoking_related_health_issues" = '1'

UNION ALL

SELECT 'Asthma' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Asthma" = '1'

UNION ALL

SELECT 'Low BP' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_Others" = '1' AND "Any_Chronic_Illnes_If_Other_specify" = 'Low BP'

UNION ALL

SELECT 'Body issues' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_Others" = '1' AND "Any_Chronic_Illnes_If_Other_specify" = 'Body issues'

UNION ALL

SELECT 'Thyroid disease' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_Others" = '1' AND "Any_Chronic_Illnes_If_Other_specify" = 'Thyroid disease'

UNION ALL

SELECT 'Thyroid' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_Others" = '1' AND "Any_Chronic_Illnes_If_Other_specify" = 'Thyroid'

UNION ALL

SELECT 'Migraine' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_Others" = '1' AND "Any_Chronic_Illnes_If_Other_specify" = 'Migraine'

UNION ALL

-- Diseases from Household_members table

SELECT 'Tuberculosis' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Tuberculosis" = '1'

UNION ALL

SELECT 'Arthritis/Back Pain' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Arthritis/Back_Pain" = '1'

UNION ALL

SELECT 'Diabetes' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Diabetes" = '1'

UNION ALL

SELECT 'Paralysis' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Paralysis" = '1'

UNION ALL

SELECT 'High BP' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/High_BP" = '1'

UNION ALL

SELECT 'Heart Disease and stroke' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Heart_Disease_and_stroke" = '1'

UNION ALL

SELECT 'Cancer' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Cancer" = '1'

UNION ALL

SELECT 'Alzheimer's Disease' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Alzheimer's_Disease" = '1'

UNION ALL

SELECT 'Kidney Diseases' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Kidney_Diseases" = '1'

UNION ALL

SELECT 'Liver Cirrhosis' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Liver_Cirrhosis" = '1'

UNION ALL

```

SELECT 'Peptic Ulcer (stomach)' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Peptic_Ulcer(stomach)" = '1'

UNION ALL

SELECT 'Alcohol related health issues' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Alcohol_related_health_issues" = '1'

UNION ALL

SELECT 'Smoking related health issues' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Smoking_related_health_issues" = '1'

UNION ALL

SELECT 'Asthma' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Asthma" = '1'

UNION ALL

SELECT 'Stomach operation' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Others" = '1' AND
"Any_Chronic_Illness>If_Other_specify," = 'Stomach_operation'

UNION ALL

SELECT 'Heartburn from oily food' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Others" =
'1' AND "Any_Chronic_Illness>If_Other_specify," = 'Heartburn_from_oily_food'
)

SELECT
Disease,
SUM(Count) AS TotalCount,
ROUND(CAST(SUM(Count) AS REAL) * 100 / (SELECT SUM(Count) FROM AllDiseases), 2) AS Percentage
FROM AllDiseases
WHERE Disease IS NOT NULL AND TotalCount > 0
GROUP BY Disease
ORDER BY TotalCount DESC;

*****sqlite

WITH AllDiseases AS (
-- Diseases from Respondent table

SELECT 'Tuberculosis' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness_/Tuberculosis" = '1'

UNION ALL

SELECT 'Arthritis/Back Pain' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness_/Arthriti_/Back_Pain" = '1'

UNION ALL

SELECT 'Diabetes' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness_/Diabetes" = '1'

UNION ALL

SELECT 'Paralysis' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness_/Paralysis" = '1'

UNION ALL

SELECT 'High BP' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness_/High_BP" = '1'

UNION ALL

SELECT 'Heart Disease and stroke' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE
"Any_Chronic_Illness_/Heart_Disease_and_stroke" = '1'

UNION ALL

SELECT 'Cancer' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness_/Cancer" = '1'

```

UNION ALL

SELECT 'Alzheimer's Disease' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness_Alzheimer's_Disease" = '1'

UNION ALL

SELECT 'Kidney Diseases' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness_Kidney_Diseases" = '1'

UNION ALL

SELECT 'Liver Cirrhosis' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness_Liver_Cirrhosis" = '1'

UNION ALL

SELECT 'Peptic Ulcer (stomach)' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness_Peptic_Ulcer(stomach)" = '1'

UNION ALL

SELECT 'Alcohol related health issues' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness_Alcohol_related_health_issues" = '1'

UNION ALL

SELECT 'Smoking related health issues' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness_Smoking_related_health_issues" = '1'

UNION ALL

SELECT 'Asthma' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness_Asthma" = '1'

UNION ALL

SELECT 'Low BP' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness_Others" = '1' AND "Any_Chronic_Illness_If_Other_specify" = 'Low BP'

UNION ALL

SELECT 'Body issues' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness_Others" = '1' AND "Any_Chronic_Illness_If_Other_specify" = 'Body issues'

UNION ALL

SELECT 'Thyroid disease' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness_Others" = '1' AND "Any_Chronic_Illness_If_Other_specify" = 'Thyroid disease'

UNION ALL

SELECT 'Thyroid' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness_Others" = '1' AND "Any_Chronic_Illness_If_Other_specify" = 'Thyroid'

UNION ALL

SELECT 'Migraine' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness_Others" = '1' AND "Any_Chronic_Illness_If_Other_specify" = 'Migraine'

UNION ALL

-- Diseases from Household_members table

SELECT 'Tuberculosis' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness_Tuberculosis" = '1'

UNION ALL

SELECT 'Arthritis/Back Pain' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness_Arthritis/Back_Pain" = '1'

UNION ALL

SELECT 'Diabetes' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness_Diabetes" = '1'

```

UNION ALL

SELECT 'Paralysis' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Paralysis" = '1'

UNION ALL

SELECT 'High BP' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/High_BP" = '1'

UNION ALL

SELECT 'Heart Disease and stroke' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Heart_Disease_and_stroke" = '1'

UNION ALL

SELECT 'Cancer' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Cancer" = '1'

UNION ALL

SELECT 'Alzheimer's Disease' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Alzheimer's_Disease" = '1'

UNION ALL

SELECT 'Kidney Diseases' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Kidney_Diseases" =
'1'

UNION ALL

SELECT 'Liver Cirrhosis' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Liver_Cirrhosis" = '1'

UNION ALL

SELECT 'Peptic Ulcer (stomach)' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Peptic_Ulcer(stomach)" = '1'

UNION ALL

SELECT 'Alcohol related health issues' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Alcohol_related_health_issues" = '1'

UNION ALL

SELECT 'Smoking related health issues' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Smoking_related_health_issues" = '1'

UNION ALL

SELECT 'Asthma' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Asthma" = '1'

UNION ALL

SELECT 'Stomach operation' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Others" = '1' AND
"Any_Chronic_Illness_If_Other_specify;" = 'Stomach_operation'

UNION ALL

SELECT 'Heartburn from oily food' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Others" =
'1' AND "Any_Chronic_Illness_If_Other_specify;" = 'Heartburn_from_oily_food'
)

SELECT
Disease,
SUM(Count) AS TotalCount,
ROUND(CAST(SUM(Count) AS REAL) * 100 / (SELECT SUM(Count) FROM AllDiseases), 2) AS Percentage
FROM AllDiseases
WHERE Disease IS NOT NULL AND TotalCount > 0
GROUP BY Disease
ORDER BY TotalCount DESC;

```

```The query is syntactically correct and appears to address the problem statement. I will now execute the query.

Action: sql\_db\_query

Action Input: ```sqlite

WITH AllDiseases AS (

-- Diseases from Respondent table

SELECT 'Tuberculosis' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Tuberculosis" = '1'

UNION ALL

SELECT 'Arthritis/Back Pain' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Arthriti\_/Back\_Pain" = '1'

UNION ALL

SELECT 'Diabetes' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Diabetes" = '1'

UNION ALL

SELECT 'Paralysis' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Paralysis" = '1'

UNION ALL

SELECT 'High BP' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/High\_BP" = '1'

UNION ALL

SELECT 'Heart Disease and stroke' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE  
"Any\_Chronic\_Illnes\_/Heart\_Disease\_and\_stroke" = '1'

UNION ALL

SELECT 'Cancer' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Cancer" = '1'

UNION ALL

SELECT 'Alzheimer's Disease' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Alzheimer's\_Disease" =  
'1'

UNION ALL

SELECT 'Kidney Diseases' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Kidney\_Diseases" = '1'

UNION ALL

SELECT 'Liver Cirrhosis' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Liver\_Cirrhosis" = '1'

UNION ALL

SELECT 'Peptic Ulcer (stomach)' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE  
"Any\_Chronic\_Illnes\_/Peptic\_Ulcer(stomach)" = '1'

UNION ALL

SELECT 'Alcohol related health issues' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE  
"Any\_Chronic\_Illnes\_/Alcohol\_related\_health\_issues" = '1'

UNION ALL

SELECT 'Smoking related health issues' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE  
"Any\_Chronic\_Illnes\_/Smoking\_related\_health\_issues" = '1'

UNION ALL

SELECT 'Asthma' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Asthma" = '1'

UNION ALL

SELECT 'Low BP' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Others" = '1' AND  
"Any\_Chronic\_Illnes\_/If\_Other\_\_specify" = 'Low BP'

UNION ALL

SELECT 'Body issues' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_Others" = '1' AND "Any\_Chronic\_Illnes\_If\_Other\_specify" = 'Body issues'

UNION ALL

SELECT 'Thyroid disease' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_Others" = '1' AND "Any\_Chronic\_Illnes\_If\_Other\_specify" = 'Thyroid disease'

UNION ALL

SELECT 'Thyroid' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_Others" = '1' AND "Any\_Chronic\_Illnes\_If\_Other\_specify" = 'Thyroid'

UNION ALL

SELECT 'Migraine' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_Others" = '1' AND "Any\_Chronic\_Illnes\_If\_Other\_specify" = 'Migraine'

UNION ALL

-- Diseases from Household\_members table

SELECT 'Tuberculosis' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE "Any\_Chronic\_Illness/Tuberculosis" = '1'

UNION ALL

SELECT 'Arthritis/Back Pain' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE "Any\_Chronic\_Illness/Arthritis/Back\_Pain" = '1'

UNION ALL

SELECT 'Diabetes' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE "Any\_Chronic\_Illness/Diabetes" = '1'

UNION ALL

SELECT 'Paralysis' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE "Any\_Chronic\_Illness/Paralysis" = '1'

UNION ALL

SELECT 'High BP' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE "Any\_Chronic\_Illness/High\_BP" = '1'

UNION ALL

SELECT 'Heart Disease and stroke' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE "Any\_Chronic\_Illness/Heart\_Disease\_and\_stroke" = '1'

UNION ALL

SELECT 'Cancer' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE "Any\_Chronic\_Illness/Cancer" = '1'

UNION ALL

SELECT 'Alzheimer's Disease' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE "Any\_Chronic\_Illness/Alzheimer's\_Disease" = '1'

UNION ALL

SELECT 'Kidney Diseases' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE "Any\_Chronic\_Illness/Kidney\_Diseases" = '1'

UNION ALL

SELECT 'Liver Cirrhosis' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE "Any\_Chronic\_Illness/Liver\_Cirrhosis" = '1'

UNION ALL

SELECT 'Peptic Ulcer (stomach)' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE "Any\_Chronic\_Illness/Peptic\_Ulcer(stomach)" = '1'

UNION ALL

```

SELECT 'Alcohol related health issues' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Alcohol_related_health_issues" = '1'

UNION ALL

SELECT 'Smoking related health issues' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Smoking_related_health_issues" = '1'

UNION ALL

SELECT 'Asthma' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Asthma" = '1'

UNION ALL

SELECT 'Stomach operation' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Others" = '1' AND
"Any_Chronic_Illness>If_Other_specify," = 'Stomach_operation'

UNION ALL

SELECT 'Heartburn from oily food' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Others" =
'1' AND "Any_Chronic_Illness>If_Other_specify," = 'Heartburn_from_oily_food'
)
SELECT
Disease,
SUM(Count) AS TotalCount,
ROUND(CAST(SUM(Count) AS REAL) * 100 / (SELECT SUM(Count) FROM AllDiseases), 2) AS Percentage
FROM AllDiseases
WHERE Disease IS NOT NULL AND TotalCount > 0
GROUP BY Disease
ORDER BY TotalCount DESC;
```


```Error: (sqlite3.OperationalError) near ""```sqlite



```

WITH AllDiseases AS (
-- Diseases from Respondent table
SELECT 'Tuberculosis' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness/Tuberculosis" = '1'

UNION ALL

SELECT 'Arthritis/Back Pain' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness/Arthritis/Back_Pain" = '1'

UNION ALL

SELECT 'Diabetes' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness/Diabetes" = '1'

UNION ALL

SELECT 'Paralysis' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness/Paralysis" = '1'

UNION ALL

SELECT 'High BP' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness/High_BP" = '1'

UNION ALL

SELECT 'Heart Disease and stroke' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE
"Any_Chronic_Illness/Heart_Disease_and_stroke" = '1'

UNION ALL

SELECT 'Cancer' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness/Cancer" = '1'

UNION ALL

SELECT 'Alzheimer's Disease' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness/Alzheimer's_Disease" =
'1'

```


```

UNION ALL

SELECT 'Kidney Diseases' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Kidney\_Diseases" = '1'

UNION ALL

SELECT 'Liver Cirrhosis' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Liver\_Cirrhosis" = '1'

UNION ALL

SELECT 'Peptic Ulcer (stomach)' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE  
"Any\_Chronic\_Illnes\_/Peptic\_Ulcer(stomach)" = '1'

UNION ALL

SELECT 'Alcohol related health issues' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE  
"Any\_Chronic\_Illnes\_/Alcohol\_related\_health\_issues" = '1'

UNION ALL

SELECT 'Smoking related health issues' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE  
"Any\_Chronic\_Illnes\_/Smoking\_related\_health\_issues" = '1'

UNION ALL

SELECT 'Asthma' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Asthma" = '1'

UNION ALL

SELECT 'Low BP' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Others" = '1' AND  
"Any\_Chronic\_Illnes\_/If\_Other\_\_\_specify" = 'Low BP'

UNION ALL

SELECT 'Body issues' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Others" = '1' AND  
"Any\_Chronic\_Illnes\_/If\_Other\_\_\_specify" = 'Body issues'

UNION ALL

SELECT 'Thyroid disease' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Others" = '1' AND  
"Any\_Chronic\_Illnes\_/If\_Other\_\_\_specify" = 'Thyroid disease'

UNION ALL

SELECT 'Thyroid' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Others" = '1' AND  
"Any\_Chronic\_Illnes\_/If\_Other\_\_\_specify" = 'Thyroid'

UNION ALL

SELECT 'Migraine' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Others" = '1' AND  
"Any\_Chronic\_Illnes\_/If\_Other\_\_\_specify" = 'Migraine'

UNION ALL

-- Diseases from Household\_members table

SELECT 'Tuberculosis' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE "Any\_Chronic\_Illness/Tuberculosis" = '1'

UNION ALL

SELECT 'Arthritis/Back Pain' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE  
"Any\_Chronic\_Illness/Arthritis/Back\_Pain" = '1'

UNION ALL

SELECT 'Diabetes' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE "Any\_Chronic\_Illness/Diabetes" = '1'

UNION ALL

SELECT 'Paralysis' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE "Any\_Chronic\_Illness/Paralysis" = '1'

UNION ALL

```

SELECT 'High BP' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/High_BP" = '1'

UNION ALL

SELECT 'Heart Disease and stroke' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Heart_Disease_and_stroke" = '1'

UNION ALL

SELECT 'Cancer' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Cancer" = '1'

UNION ALL

SELECT 'Alzheimer's Disease' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Alzheimer's_Disease" = '1'

UNION ALL

SELECT 'Kidney Diseases' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Kidney_Diseases" =
'1'

UNION ALL

SELECT 'Liver Cirrhosis' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Liver_Cirrhosis" = '1'

UNION ALL

SELECT 'Peptic Ulcer (stomach)' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Peptic_Ulcer(stomach)" = '1'

UNION ALL

SELECT 'Alcohol related health issues' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Alcohol_related_health_issues" = '1'

UNION ALL

SELECT 'Smoking related health issues' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Smoking_related_health_issues" = '1'

UNION ALL

SELECT 'Asthma' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Asthma" = '1'

UNION ALL

SELECT 'Stomach operation' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Others" = '1' AND
"Any_Chronic_Illness_If_Other_specify" = 'Stomach_operation'

UNION ALL

SELECT 'Heartburn from oily food' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Others" =
'1' AND "Any_Chronic_Illness_If_Other_specify" = 'Heartburn_from_oily_food'

)

SELECT
Disease,
SUM(Count) AS TotalCount,
ROUND(CAST(SUM(Count) AS REAL) * 100 / (SELECT SUM(Count) FROM AllDiseases), 2) AS Percentage
FROM AllDiseases
WHERE Disease IS NOT NULL AND TotalCount > 0
GROUP BY Disease
ORDER BY TotalCount DESC;
***: syntax error
[SQL: ``sqlite
WITH AllDiseases AS (

```

-- Diseases from Respondent table

SELECT 'Tuberculosis' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Tuberculosis" = '1'

UNION ALL

SELECT 'Arthritis/Back Pain' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Arthriti\_/Back\_Pain" = '1'

UNION ALL

SELECT 'Diabetes' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Diabetes" = '1'

UNION ALL

SELECT 'Paralysis' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Paralysis" = '1'

UNION ALL

SELECT 'High BP' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/High\_BP" = '1'

UNION ALL

SELECT 'Heart Disease and stroke' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE  
"Any\_Chronic\_Illnes\_/Heart\_Disease\_and\_stroke" = '1'

UNION ALL

SELECT 'Cancer' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Cancer" = '1'

UNION ALL

SELECT 'Alzheimer's Disease' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Alzheimer's\_Disease" =  
'1'

UNION ALL

SELECT 'Kidney Diseases' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Kidney\_Diseases" = '1'

UNION ALL

SELECT 'Liver Cirrhosis' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Liver\_Cirrhosis" = '1'

UNION ALL

SELECT 'Peptic Ulcer (stomach)' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE  
"Any\_Chronic\_Illnes\_/Peptic\_Ulcer(stomach)" = '1'

UNION ALL

SELECT 'Alcohol related health issues' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE  
"Any\_Chronic\_Illnes\_/Alcohol\_related\_health\_issues" = '1'

UNION ALL

SELECT 'Smoking related health issues' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE  
"Any\_Chronic\_Illnes\_/Smoking\_related\_health\_issues" = '1'

UNION ALL

SELECT 'Asthma' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Asthma" = '1'

UNION ALL

SELECT 'Low BP' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Others" = '1' AND  
"Any\_Chronic\_Illnes\_/If\_Other\_\_specify" = 'Low BP'

UNION ALL

SELECT 'Body issues' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Others" = '1' AND  
"Any\_Chronic\_Illnes\_/If\_Other\_\_specify" = 'Body issues'

UNION ALL

SELECT 'Thyroid disease' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Others" = '1' AND  
"Any\_Chronic\_Illnes\_/If\_Other\_\_specify" = 'Thyroid disease'

UNION ALL

```
SELECT 'Thyroid' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness_Others" = '1' AND
"Any_Chronic_Illness_If_Other_specify" = 'Thyroid'
```

UNION ALL

```
SELECT 'Migraine' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness_Others" = '1' AND
"Any_Chronic_Illness_If_Other_specify" = 'Migraine'
```

UNION ALL

-- Diseases from Household\_members table

```
SELECT 'Tuberculosis' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Tuberculosis" = '1'
```

UNION ALL

```
SELECT 'Arthritis/Back Pain' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Arthritis/Back_Pain" = '1'
```

UNION ALL

```
SELECT 'Diabetes' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Diabetes" = '1'
```

UNION ALL

```
SELECT 'Paralysis' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Paralysis" = '1'
```

UNION ALL

```
SELECT 'High BP' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/High_BP" = '1'
```

UNION ALL

```
SELECT 'Heart Disease and stroke' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Heart_Disease_and_stroke" = '1'
```

UNION ALL

```
SELECT 'Cancer' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Cancer" = '1'
```

UNION ALL

```
SELECT 'Alzheimer's Disease' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Alzheimer's_Disease" = '1'
```

UNION ALL

```
SELECT 'Kidney Diseases' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Kidney_Diseases" =
'1'
```

UNION ALL

```
SELECT 'Liver Cirrhosis' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Liver_Cirrhosis" = '1'
```

UNION ALL

```
SELECT 'Peptic Ulcer (stomach)' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Peptic_Ulcer(stomach)" = '1'
```

UNION ALL

```
SELECT 'Alcohol related health issues' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Alcohol_related_health_issues" = '1'
```

UNION ALL

```
SELECT 'Smoking related health issues' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Smoking_related_health_issues" = '1'
```

UNION ALL

```
SELECT 'Asthma' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Asthma" = '1'
```

```

UNION ALL

SELECT 'Stomach operation' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Others" = '1' AND
"Any_Chronic_Illness_If_Other_specify" = 'Stomach_operation'

UNION ALL

SELECT 'Heartburn from oily food' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Others" =
'1' AND "Any_Chronic_Illness_If_Other_specify" = 'Heartburn_from_oily_food'
)
SELECT
Disease,
SUM(Count) AS TotalCount,
ROUND(CAST(SUM(Count) AS REAL) * 100 / (SELECT SUM(Count) FROM AllDiseases), 2) AS Percentage
FROM AllDiseases
WHERE Disease IS NOT NULL AND TotalCount > 0
GROUP BY Disease
ORDER BY TotalCount DESC;
'''
(Background on this error at: https://sqlalche.me/e/20/e3q8)Action: sql_db_query_checker
Action Input: '''
WITH AllDiseases AS (
-- Diseases from Respondent table
SELECT 'Tuberculosis' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_JTuberculosis" = '1'
UNION ALL
SELECT 'Arthritis/Back Pain' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_JArthriti_Back_Pain" = '1'
UNION ALL
SELECT 'Diabetes' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_JDiabetes" = '1'
UNION ALL
SELECT 'Paralysis' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_JParalysis" = '1'
UNION ALL
SELECT 'High BP' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_JHigh_BP" = '1'
UNION ALL
SELECT 'Heart Disease and stroke' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE
"Any_Chronic_Illnes_JHeart_Disease_and_stroke" = '1'
UNION ALL
SELECT 'Cancer' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_JCancer" = '1'
UNION ALL
SELECT 'Alzheimer's Disease' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_JAlzheimer's_Disease" =
'1'
UNION ALL
SELECT 'Kidney Diseases' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_JKidney_Diseases" = '1'
UNION ALL
SELECT 'Liver Cirrhosis' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_JLiver_Cirrhosis" = '1'

```

UNION ALL

SELECT 'Peptic Ulcer (stomach)' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE  
"Any\_Chronic\_Illnes\_/Peptic\_Ulcer(stomach)" = '1'

UNION ALL

SELECT 'Alcohol related health issues' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE  
"Any\_Chronic\_Illnes\_/Alcohol\_related\_health\_issues" = '1'

UNION ALL

SELECT 'Smoking related health issues' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE  
"Any\_Chronic\_Illnes\_/Smoking\_related\_health\_issues" = '1'

UNION ALL

SELECT 'Asthma' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Asthma" = '1'

UNION ALL

SELECT 'Low BP' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Others" = '1' AND  
"Any\_Chronic\_Illnes\_/If\_Other\_\_\_specify" = 'Low BP'

UNION ALL

SELECT 'Body issues' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Others" = '1' AND  
"Any\_Chronic\_Illnes\_/If\_Other\_\_\_specify" = 'Body issues'

UNION ALL

SELECT 'Thyroid disease' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Others" = '1' AND  
"Any\_Chronic\_Illnes\_/If\_Other\_\_\_specify" = 'Thyroid disease'

UNION ALL

SELECT 'Thyroid' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Others" = '1' AND  
"Any\_Chronic\_Illnes\_/If\_Other\_\_\_specify" = 'Thyroid'

UNION ALL

SELECT 'Migraine' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Others" = '1' AND  
"Any\_Chronic\_Illnes\_/If\_Other\_\_\_specify" = 'Migraine'

UNION ALL

-- Diseases from Household\_members table

SELECT 'Tuberculosis' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE "Any\_Chronic\_Illness/Tuberculosis" = '1'

UNION ALL

SELECT 'Arthritis/Back Pain' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE  
"Any\_Chronic\_Illness/Arthritis/Back\_Pain" = '1'

UNION ALL

SELECT 'Diabetes' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE "Any\_Chronic\_Illness/Diabetes" = '1'

UNION ALL

SELECT 'Paralysis' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE "Any\_Chronic\_Illness/Paralysis" = '1'

UNION ALL

SELECT 'High BP' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE "Any\_Chronic\_Illness/High\_BP" = '1'

UNION ALL

SELECT 'Heart Disease and stroke' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE  
"Any\_Chronic\_Illness/Heart\_Disease\_and\_stroke" = '1'

```

UNION ALL

SELECT 'Cancer' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Cancer" = '1'

UNION ALL

SELECT 'Alzheimer's Disease' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Alzheimer's_Disease" = '1'

UNION ALL

SELECT 'Kidney Diseases' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Kidney_Diseases" =
'1'

UNION ALL

SELECT 'Liver Cirrhosis' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Liver_Cirrhosis" = '1'

UNION ALL

SELECT 'Peptic Ulcer (stomach)' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Peptic_Ulcer(stomach)" = '1'

UNION ALL

SELECT 'Alcohol related health issues' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Alcohol_related_health_issues" = '1'

UNION ALL

SELECT 'Smoking related health issues' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Smoking_related_health_issues" = '1'

UNION ALL

SELECT 'Asthma' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Asthma" = '1'

UNION ALL

SELECT 'Stomach operation' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Others" = '1' AND
"Any_Chronic_Illness_If_Other_specify" = 'Stomach_operation'

UNION ALL

SELECT 'Heartburn from oily food' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Others" =
'1' AND "Any_Chronic_Illness_If_Other_specify" = 'Heartburn_from_oily_food'
)
SELECT
Disease,
SUM(Count) AS TotalCount,
ROUND(CAST(SUM(Count) AS REAL) * 100 / (SELECT SUM(Count) FROM AllDiseases), 2) AS Percentage
FROM AllDiseases
WHERE Disease IS NOT NULL AND TotalCount > 0
GROUP BY Disease
ORDER BY TotalCount DESC;
*****sqlite
WITH AllDiseases AS (
-- Diseases from Respondent table
SELECT 'Tuberculosis' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness/Tuberculosis" = '1'

UNION ALL

SELECT 'Arthritis/Back Pain' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness/Arthriti_/Back_Pain" = '1'

```

UNION ALL

SELECT 'Diabetes' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Diabetes" = '1'

UNION ALL

SELECT 'Paralysis' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Paralysis" = '1'

UNION ALL

SELECT 'High BP' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/High\_BP" = '1'

UNION ALL

SELECT 'Heart Disease and stroke' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Heart\_Disease\_and\_stroke" = '1'

UNION ALL

SELECT 'Cancer' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Cancer" = '1'

UNION ALL

SELECT 'Alzheimer's Disease' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Alzheimer's\_Disease" = '1'

UNION ALL

SELECT 'Kidney Diseases' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Kidney\_Diseases" = '1'

UNION ALL

SELECT 'Liver Cirrhosis' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Liver\_Cirrhosis" = '1'

UNION ALL

SELECT 'Peptic Ulcer (stomach)' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Peptic\_Ulcer(stomach)" = '1'

UNION ALL

SELECT 'Alcohol related health issues' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Alcohol\_related\_health\_issues" = '1'

UNION ALL

SELECT 'Smoking related health issues' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Smoking\_related\_health\_issues" = '1'

UNION ALL

SELECT 'Asthma' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Asthma" = '1'

UNION ALL

SELECT 'Low BP' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Others" = '1' AND "Any\_Chronic\_Illnes\_/If\_Other\_\_specify" = 'Low BP'

UNION ALL

SELECT 'Body issues' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Others" = '1' AND "Any\_Chronic\_Illnes\_/If\_Other\_\_specify" = 'Body issues'

UNION ALL

SELECT 'Thyroid disease' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Others" = '1' AND "Any\_Chronic\_Illnes\_/If\_Other\_\_specify" = 'Thyroid disease'

UNION ALL

SELECT 'Thyroid' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Others" = '1' AND "Any\_Chronic\_Illnes\_/If\_Other\_\_specify" = 'Thyroid'

UNION ALL

```
SELECT 'Migraine' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Others" = '1' AND
"Any_Chronic_Illnes_/If_Other_specify" = 'Migraine'
```

UNION ALL

-- Diseases from Household\_members table

```
SELECT 'Tuberculosis' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Tuberculosis" = '1'
```

UNION ALL

```
SELECT 'Arthritis/Back Pain' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Arthritis/Back_Pain" = '1'
```

UNION ALL

```
SELECT 'Diabetes' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Diabetes" = '1'
```

UNION ALL

```
SELECT 'Paralysis' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Paralysis" = '1'
```

UNION ALL

```
SELECT 'High BP' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/High_BP" = '1'
```

UNION ALL

```
SELECT 'Heart Disease and stroke' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Heart_Disease_and_stroke" = '1'
```

UNION ALL

```
SELECT 'Cancer' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Cancer" = '1'
```

UNION ALL

```
SELECT 'Alzheimer's Disease' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Alzheimer's_Disease" = '1'
```

UNION ALL

```
SELECT 'Kidney Diseases' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Kidney_Diseases" =
'1'
```

UNION ALL

```
SELECT 'Liver Cirrhosis' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Liver_Cirrhosis" = '1'
```

UNION ALL

```
SELECT 'Peptic Ulcer (stomach)' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Peptic_Ulcer(stomach)" = '1'
```

UNION ALL

```
SELECT 'Alcohol related health issues' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Alcohol_related_health_issues" = '1'
```

UNION ALL

```
SELECT 'Smoking related health issues' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Smoking_related_health_issues" = '1'
```

UNION ALL

```
SELECT 'Asthma' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Asthma" = '1'
```

UNION ALL

```
SELECT 'Stomach operation' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Others" = '1' AND
"Any_Chronic_Illness_/If_Other_specify;" = 'Stomach_operation'
```

UNION ALL

```
SELECT 'Heartburn from oily food' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Others" = '1' AND "Any_Chronic_Illness_if_Other_specify" = 'Heartburn_from_oily_food'
```

)

SELECT

Disease,

SUM(Count) AS TotalCount,

ROUND(CAST(SUM(Count) AS REAL) \* 100 / (SELECT SUM(Count) FROM AllDiseases), 2) AS Percentage

FROM AllDiseases

WHERE Disease IS NOT NULL AND TotalCount > 0

GROUP BY Disease

ORDER BY TotalCount DESC;

``thought: The previous `sql\_db\_query` failed due to a syntax error, likely caused by the triple backticks being included in the `Action Input` itself, which the tool interprets as part of the SQL query. I need to remove the triple backticks from the `Action Input` when calling `sql\_db\_query`. The query itself was validated as correct by `sql\_db\_query\_checker`.

I will re-execute the query without the surrounding backticks.Action: sql\_db\_query

Action Input: WITH AllDiseases AS (

-- Diseases from Respondent table

```
SELECT 'Tuberculosis' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Tuberculosis" = '1'
```

UNION ALL

```
SELECT 'Arthritis/Back Pain' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Arthriti_/Back_Pain" = '1'
```

UNION ALL

```
SELECT 'Diabetes' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Diabetes" = '1'
```

UNION ALL

```
SELECT 'Paralysis' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Paralysis" = '1'
```

UNION ALL

```
SELECT 'High BP' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/High_BP" = '1'
```

UNION ALL

```
SELECT 'Heart Disease and stroke' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Heart_Disease_and_stroke" = '1'
```

UNION ALL

```
SELECT 'Cancer' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Cancer" = '1'
```

UNION ALL

```
SELECT 'Alzheimer's Disease' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Alzheimer's_Disease" = '1'
```

UNION ALL

```
SELECT 'Kidney Diseases' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Kidney_Diseases" = '1'
```

UNION ALL

```
SELECT 'Liver Cirrhosis' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Liver_Cirrhosis" = '1'
```

UNION ALL

SELECT 'Peptic Ulcer (stomach)' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE  
"Any\_Chronic\_Illnes\_/Peptic\_Ulcer(stomach)" = '1'

UNION ALL

SELECT 'Alcohol related health issues' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE  
"Any\_Chronic\_Illnes\_/Alcohol\_related\_health\_issues" = '1'

UNION ALL

SELECT 'Smoking related health issues' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE  
"Any\_Chronic\_Illnes\_/Smoking\_related\_health\_issues" = '1'

UNION ALL

SELECT 'Asthma' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Asthma" = '1'

UNION ALL

SELECT 'Low BP' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Others" = '1' AND  
"Any\_Chronic\_Illnes\_/If\_Other\_\_\_specify" = 'Low BP'

UNION ALL

SELECT 'Body issues' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Others" = '1' AND  
"Any\_Chronic\_Illnes\_/If\_Other\_\_\_specify" = 'Body issues'

UNION ALL

SELECT 'Thyroid disease' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Others" = '1' AND  
"Any\_Chronic\_Illnes\_/If\_Other\_\_\_specify" = 'Thyroid disease'

UNION ALL

SELECT 'Thyroid' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Others" = '1' AND  
"Any\_Chronic\_Illnes\_/If\_Other\_\_\_specify" = 'Thyroid'

UNION ALL

SELECT 'Migraine' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Others" = '1' AND  
"Any\_Chronic\_Illnes\_/If\_Other\_\_\_specify" = 'Migraine'

UNION ALL

-- Diseases from Household\_members table

SELECT 'Tuberculosis' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE "Any\_Chronic\_Illness/Tuberculosis" = '1'

UNION ALL

SELECT 'Arthritis/Back Pain' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE  
"Any\_Chronic\_Illness/Arthritis/Back\_Pain" = '1'

UNION ALL

SELECT 'Diabetes' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE "Any\_Chronic\_Illness/Diabetes" = '1'

UNION ALL

SELECT 'Paralysis' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE "Any\_Chronic\_Illness/Paralysis" = '1'

UNION ALL

SELECT 'High BP' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE "Any\_Chronic\_Illness/High\_BP" = '1'

UNION ALL

SELECT 'Heart Disease and stroke' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE  
"Any\_Chronic\_Illness/Heart\_Disease\_and\_stroke" = '1'

UNION ALL

```

SELECT 'Cancer' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Cancer" = '1'

UNION ALL

SELECT 'Alzheimer's Disease' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Alzheimer"s_Disease" = '1'

UNION ALL

SELECT 'Kidney Diseases' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Kidney_Diseases" =
'1'

UNION ALL

SELECT 'Liver Cirrhosis' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Liver_Cirrhosis" = '1'

UNION ALL

SELECT 'Peptic Ulcer (stomach)' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Peptic_Ulcer(stomach)" = '1'

UNION ALL

SELECT 'Alcohol related health issues' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Alcohol_related_health_issues" = '1'

UNION ALL

SELECT 'Smoking related health issues' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Smoking_related_health_issues" = '1'

UNION ALL

SELECT 'Asthma' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Asthma" = '1'

UNION ALL

SELECT 'Stomach operation' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Others" = '1' AND
"Any_Chronic_Illness_If_Other_specify" = 'Stomach_operation'

UNION ALL

SELECT 'Heartburn from oily food' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Others" =
'1' AND "Any_Chronic_Illness_If_Other_specify" = 'Heartburn_from_oily_food'
)
SELECT
Disease,
SUM(Count) AS TotalCount,
ROUND(CAST(SUM(Count) AS REAL) * 100 / (SELECT SUM(Count) FROM AllDiseases), 2) AS Percentage
FROM AllDiseases
WHERE Disease IS NOT NULL AND TotalCount > 0
GROUP BY Disease
ORDER BY TotalCount DESC;
0 / (SELECT SUM(Count) FROM AllDiseases), 2) AS Percentage
FROM AllDiseases
WHERE Disease IS NOT NULL AND TotalCount > 0
GROUP BY Disease
ORDER BY TotalCount DESC;
Error: (sqlite3.OperationalError) misuse of aggregate: SUM()
[SQL: WITH AllDiseases AS (

```

-- Diseases from Respondent table

SELECT 'Tuberculosis' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Tuberculosis" = '1'

UNION ALL

SELECT 'Arthritis/Back Pain' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Arthriti\_/Back\_Pain" = '1'

UNION ALL

SELECT 'Diabetes' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Diabetes" = '1'

UNION ALL

SELECT 'Paralysis' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Paralysis" = '1'

UNION ALL

SELECT 'High BP' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/High\_BP" = '1'

UNION ALL

SELECT 'Heart Disease and stroke' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE  
"Any\_Chronic\_Illnes\_/Heart\_Disease\_and\_stroke" = '1'

UNION ALL

SELECT 'Cancer' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Cancer" = '1'

UNION ALL

SELECT 'Alzheimer's Disease' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Alzheimer's\_Disease" =  
'1'

UNION ALL

SELECT 'Kidney Diseases' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Kidney\_Diseases" = '1'

UNION ALL

SELECT 'Liver Cirrhosis' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Liver\_Cirrhosis" = '1'

UNION ALL

SELECT 'Peptic Ulcer (stomach)' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE  
"Any\_Chronic\_Illnes\_/Peptic\_Ulcer(stomach)" = '1'

UNION ALL

SELECT 'Alcohol related health issues' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE  
"Any\_Chronic\_Illnes\_/Alcohol\_related\_health\_issues" = '1'

UNION ALL

SELECT 'Smoking related health issues' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE  
"Any\_Chronic\_Illnes\_/Smoking\_related\_health\_issues" = '1'

UNION ALL

SELECT 'Asthma' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Asthma" = '1'

UNION ALL

SELECT 'Low BP' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Others" = '1' AND  
"Any\_Chronic\_Illnes\_/If\_Other\_specify" = 'Low BP'

UNION ALL

SELECT 'Body issues' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Others" = '1' AND  
"Any\_Chronic\_Illnes\_/If\_Other\_specify" = 'Body issues'

UNION ALL

SELECT 'Thyroid disease' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Others" = '1' AND  
"Any\_Chronic\_Illnes\_/If\_Other\_specify" = 'Thyroid disease'

UNION ALL

```
SELECT 'Thyroid' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness_Others" = '1' AND "Any_Chronic_Illness_If_Other_specify" = 'Thyroid'
```

UNION ALL

```
SELECT 'Migraine' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness_Others" = '1' AND "Any_Chronic_Illness_If_Other_specify" = 'Migraine'
```

UNION ALL

-- Diseases from Household\_members table

```
SELECT 'Tuberculosis' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Tuberculosis" = '1'
```

UNION ALL

```
SELECT 'Arthritis/Back Pain' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Arthritis/Back_Pain" = '1'
```

UNION ALL

```
SELECT 'Diabetes' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Diabetes" = '1'
```

UNION ALL

```
SELECT 'Paralysis' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Paralysis" = '1'
```

UNION ALL

```
SELECT 'High BP' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/High_BP" = '1'
```

UNION ALL

```
SELECT 'Heart Disease and stroke' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Heart_Disease_and_stroke" = '1'
```

UNION ALL

```
SELECT 'Cancer' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Cancer" = '1'
```

UNION ALL

```
SELECT 'Alzheimer's Disease' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Alzheimer's_Disease" = '1'
```

UNION ALL

```
SELECT 'Kidney Diseases' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Kidney_Diseases" = '1'
```

UNION ALL

```
SELECT 'Liver Cirrhosis' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Liver_Cirrhosis" = '1'
```

UNION ALL

```
SELECT 'Peptic Ulcer (stomach)' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Peptic_Ulcer(stomach)" = '1'
```

UNION ALL

```
SELECT 'Alcohol related health issues' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Alcohol_related_health_issues" = '1'
```

UNION ALL

```
SELECT 'Smoking related health issues' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Smoking_related_health_issues" = '1'
```

UNION ALL

```
SELECT 'Asthma' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Asthma" = '1'
```

```

UNION ALL

SELECT 'Stomach operation' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Others" = '1' AND
"Any_Chronic_Illness_If_Other_specify" = 'Stomach_operation'

UNION ALL

SELECT 'Heartburn from oily food' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Others" =
'1' AND "Any_Chronic_Illness_If_Other_specify" = 'Heartburn_from_oily_food'
)
SELECT
Disease,
SUM(Count) AS TotalCount,
ROUND(CAST(SUM(Count) AS REAL) * 100 / (SELECT SUM(Count) FROM AllDiseases), 2) AS Percentage
FROM AllDiseases
WHERE Disease IS NOT NULL AND TotalCount > 0
GROUP BY Disease
ORDER BY TotalCount DESC;
0 / (SELECT SUM(Count) FROM AllDiseases), 2) AS Percentage
FROM AllDiseases
WHERE Disease IS NOT NULL AND TotalCount > 0
GROUP BY Disease
ORDER BY TotalCount DESC;
]

```

(Background on this error at: <https://sqlalche.me/e/20/e3q8>) The error "misuse of aggregate: SUM()" indicates that there's an issue with how 'SUM(Count)' is being used, specifically in the subquery '(SELECT SUM(Count) FROM AllDiseases)'. This usually happens when an aggregate function is used in a context where it's not allowed, or if there's a syntax error that makes the parser misinterpret the query.

Looking at the previous 'Action Input', I see a repeated part of the query at the very end:

```

0 / (SELECT SUM(Count) FROM AllDiseases), 2) AS Percentage FROM AllDiseases WHERE Disease IS NOT NULL AND TotalCount > 0
GROUP BY Disease ORDER BY TotalCount DESC;

```

This repetition is definitely a syntax error and is likely causing the "misuse of aggregate" error. I need to remove this duplicate part.

Let's correct the query by removing the duplicate part and re-run the 'sql\_db\_query\_checker' first.

Action: sql\_db\_query\_checker

Action Input: ``sqlite

```

WITH AllDiseases AS (
 -- Diseases from Respondent table
 SELECT 'Tuberculosis' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Tuberculosis" = '1'
 UNION ALL
 SELECT 'Arthritis/Back Pain' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Arthriti_/Back_Pain" = '1'
 UNION ALL
 SELECT 'Diabetes' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Diabetes" = '1'
 UNION ALL

```

```

SELECT 'Paralysis' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Paralysis" = '1'

UNION ALL

SELECT 'High BP' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/High_BP" = '1'

UNION ALL

SELECT 'Heart Disease and stroke' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE
"Any_Chronic_Illnes_/Heart_Disease_and_stroke" = '1'

UNION ALL

SELECT 'Cancer' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Cancer" = '1'

UNION ALL

SELECT 'Alzheimer's Disease' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Alzheimer's_Disease" =
'1'

UNION ALL

SELECT 'Kidney Diseases' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Kidney_Diseases" = '1'

UNION ALL

SELECT 'Liver Cirrhosis' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Liver_Cirrhosis" = '1'

UNION ALL

SELECT 'Peptic Ulcer (stomach)' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE
"Any_Chronic_Illnes_/Peptic_Ulcer(stomach)" = '1'

UNION ALL

SELECT 'Alcohol related health issues' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE
"Any_Chronic_Illnes_/Alcohol_related_health_issues" = '1'

UNION ALL

SELECT 'Smoking related health issues' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE
"Any_Chronic_Illnes_/Smoking_related_health_issues" = '1'

UNION ALL

SELECT 'Asthma' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Asthma" = '1'

UNION ALL

SELECT 'Low BP' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Others" = '1' AND
"Any_Chronic_Illnes_/If_Other___specify" = 'Low BP'

UNION ALL

SELECT 'Body issues' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Others" = '1' AND
"Any_Chronic_Illnes_/If_Other___specify" = 'Body issues'

UNION ALL

SELECT 'Thyroid disease' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Others" = '1' AND
"Any_Chronic_Illnes_/If_Other___specify" = 'Thyroid disease'

UNION ALL

SELECT 'Thyroid' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Others" = '1' AND
"Any_Chronic_Illnes_/If_Other___specify" = 'Thyroid'

UNION ALL

SELECT 'Migraine' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Others" = '1' AND
"Any_Chronic_Illnes_/If_Other___specify" = 'Migraine'

UNION ALL

```

```

-- Diseases from Household_members table

SELECT 'Tuberculosis' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Tuberculosis" = '1'

UNION ALL

SELECT 'Arthritis/Back Pain' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Arthritis/Back_Pain" = '1'

UNION ALL

SELECT 'Diabetes' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Diabetes" = '1'

UNION ALL

SELECT 'Paralysis' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Paralysis" = '1'

UNION ALL

SELECT 'High BP' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/High_BP" = '1'

UNION ALL

SELECT 'Heart Disease and stroke' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Heart_Disease_and_stroke" = '1'

UNION ALL

SELECT 'Cancer' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Cancer" = '1'

UNION ALL

SELECT 'Alzheimer's Disease' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Alzheimer's_Disease" = '1'

UNION ALL

SELECT 'Kidney Diseases' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Kidney_Diseases" =
'1'

UNION ALL

SELECT 'Liver Cirrhosis' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Liver_Cirrhosis" = '1'

UNION ALL

SELECT 'Peptic Ulcer (stomach)' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Peptic_Ulcer(stomach)" = '1'

UNION ALL

SELECT 'Alcohol related health issues' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Alcohol_related_health_issues" = '1'

UNION ALL

SELECT 'Smoking related health issues' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Smoking_related_health_issues" = '1'

UNION ALL

SELECT 'Asthma' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Asthma" = '1'

UNION ALL

SELECT 'Stomach operation' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Others" = '1' AND
"Any_Chronic_Illness_If_Other_specify." = 'Stomach_operation'

UNION ALL

SELECT 'Heartburn from oily food' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Others" =
'1' AND "Any_Chronic_Illness_If_Other_specify." = 'Heartburn_from_oily_food'

)

```

```

SELECT
 Disease,
 SUM(Count) AS TotalCount,
 ROUND(CAST(SUM(Count) AS REAL) * 100 / (SELECT SUM(Count) FROM AllDiseases), 2) AS Percentage
FROM AllDiseases
WHERE Disease IS NOT NULL AND TotalCount > 0
GROUP BY Disease
ORDER BY TotalCount DESC;
''''''sqlite
WITH AllDiseases AS (
 -- Diseases from Respondent table
 SELECT 'Tuberculosis' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Tuberculosis" = '1'
 UNION ALL
 SELECT 'Arthritis/Back Pain' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Arthriti_/Back_Pain" = '1'
 UNION ALL
 SELECT 'Diabetes' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Diabetes" = '1'
 UNION ALL
 SELECT 'Paralysis' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Paralysis" = '1'
 UNION ALL
 SELECT 'High BP' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/High_BP" = '1'
 UNION ALL
 SELECT 'Heart Disease and stroke' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE
 "Any_Chronic_Illnes_/Heart_Disease_and_stroke" = '1'
 UNION ALL
 SELECT 'Cancer' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Cancer" = '1'
 UNION ALL
 SELECT 'Alzheimer's Disease' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Alzheimer's_Disease" =
 '1'
 UNION ALL
 SELECT 'Kidney Diseases' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Kidney_Diseases" = '1'
 UNION ALL
 SELECT 'Liver Cirrhosis' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Liver_Cirrhosis" = '1'
 UNION ALL
 SELECT 'Peptic Ulcer (stomach)' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE
 "Any_Chronic_Illnes_/Peptic_Ulcer(stomach)" = '1'
 UNION ALL
 SELECT 'Alcohol related health issues' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE
 "Any_Chronic_Illnes_/Alcohol_related_health_issues" = '1'
 UNION ALL
 SELECT 'Smoking related health issues' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE
 "Any_Chronic_Illnes_/Smoking_related_health_issues" = '1'

```

UNION ALL

SELECT 'Asthma' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Asthma" = '1'

UNION ALL

SELECT 'Low BP' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Others" = '1' AND "Any\_Chronic\_Illnes\_/If\_Other\_\_\_specify" = 'Low BP'

UNION ALL

SELECT 'Body issues' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Others" = '1' AND "Any\_Chronic\_Illnes\_/If\_Other\_\_\_specify" = 'Body issues'

UNION ALL

SELECT 'Thyroid disease' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Others" = '1' AND "Any\_Chronic\_Illnes\_/If\_Other\_\_\_specify" = 'Thyroid disease'

UNION ALL

SELECT 'Thyroid' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Others" = '1' AND "Any\_Chronic\_Illnes\_/If\_Other\_\_\_specify" = 'Thyroid'

UNION ALL

SELECT 'Migraine' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Others" = '1' AND "Any\_Chronic\_Illnes\_/If\_Other\_\_\_specify" = 'Migraine'

UNION ALL

-- Diseases from Household\_members table

SELECT 'Tuberculosis' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE "Any\_Chronic\_Illness/Tuberculosis" = '1'

UNION ALL

SELECT 'Arthritis/Back Pain' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE "Any\_Chronic\_Illness/Arthritis/Back\_Pain" = '1'

UNION ALL

SELECT 'Diabetes' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE "Any\_Chronic\_Illness/Diabetes" = '1'

UNION ALL

SELECT 'Paralysis' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE "Any\_Chronic\_Illness/Paralysis" = '1'

UNION ALL

SELECT 'High BP' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE "Any\_Chronic\_Illness/High\_BP" = '1'

UNION ALL

SELECT 'Heart Disease and stroke' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE "Any\_Chronic\_Illness/Heart\_Disease\_and\_stroke" = '1'

UNION ALL

SELECT 'Cancer' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE "Any\_Chronic\_Illness/Cancer" = '1'

UNION ALL

SELECT 'Alzheimer's Disease' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE "Any\_Chronic\_Illness/Alzheimer's\_Disease" = '1'

UNION ALL

SELECT 'Kidney Diseases' AS Disease, COUNT(\*) AS Count FROM "Household\_members" WHERE "Any\_Chronic\_Illness/Kidney\_Diseases" = '1'

UNION ALL

```

SELECT 'Liver Cirrhosis' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Liver_Cirrhosis" = '1'

UNION ALL

SELECT 'Peptic Ulcer (stomach)' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Peptic_Ulcer(stomach)" = '1'

UNION ALL

SELECT 'Alcohol related health issues' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Alcohol_related_health_issues" = '1'

UNION ALL

SELECT 'Smoking related health issues' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Smoking_related_health_issues" = '1'

UNION ALL

SELECT 'Asthma' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Asthma" = '1'

UNION ALL

SELECT 'Stomach operation' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Others" = '1' AND
"Any_Chronic_Illness_If_Other_specify" = 'Stomach_operation'

UNION ALL

SELECT 'Heartburn from oily food' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Others" =
'1' AND "Any_Chronic_Illness_If_Other_specify" = 'Heartburn_from_oily_food'
)

```

SELECT

Disease,

SUM(Count) AS TotalCount,

ROUND(CAST(SUM(Count) AS REAL) \* 100 / (SELECT SUM(Count) FROM AllDiseases), 2) AS Percentage

FROM AllDiseases

GROUP BY Disease

HAVING TotalCount > 0

ORDER BY TotalCount DESC;

```The query is syntactically correct and appears to address the problem statement. I will now execute the query.

Action: sql_db_query

Action Input: WITH AllDiseases AS (

-- Diseases from Respondent table

```

SELECT 'Tuberculosis' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness_/Tuberculosis" = '1'

```

UNION ALL

```

SELECT 'Arthritis/Back Pain' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness_/Arthriti_/Back_Pain" = '1'

```

UNION ALL

```

SELECT 'Diabetes' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness_/Diabetes" = '1'

```

UNION ALL

```

SELECT 'Paralysis' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness_/Paralysis" = '1'

```

UNION ALL

```

SELECT 'High BP' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illness_/High_BP" = '1'

```

UNION ALL

```
SELECT 'Heart Disease and stroke' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE  
"Any_Chronic_Illnes_/Heart_Disease_and_stroke" = '1'
```

UNION ALL

```
SELECT 'Cancer' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Cancer" = '1'
```

UNION ALL

```
SELECT 'Alzheimer's Disease' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Alzheimer's_Disease" =  
'1'
```

UNION ALL

```
SELECT 'Kidney Diseases' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Kidney_Diseases" = '1'
```

UNION ALL

```
SELECT 'Liver Cirrhosis' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Liver_Cirrhosis" = '1'
```

UNION ALL

```
SELECT 'Peptic Ulcer (stomach)' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE  
"Any_Chronic_Illnes_/Peptic_Ulcer(stomach)" = '1'
```

UNION ALL

```
SELECT 'Alcohol related health issues' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE  
"Any_Chronic_Illnes_/Alcohol_related_health_issues" = '1'
```

UNION ALL

```
SELECT 'Smoking related health issues' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE  
"Any_Chronic_Illnes_/Smoking_related_health_issues" = '1'
```

UNION ALL

```
SELECT 'Asthma' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Asthma" = '1'
```

UNION ALL

```
SELECT 'Low BP' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Others" = '1' AND  
"Any_Chronic_Illnes_/If_Other___specify" = 'Low BP'
```

UNION ALL

```
SELECT 'Body issues' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Others" = '1' AND  
"Any_Chronic_Illnes_/If_Other___specify" = 'Body issues'
```

UNION ALL

```
SELECT 'Thyroid disease' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Others" = '1' AND  
"Any_Chronic_Illnes_/If_Other___specify" = 'Thyroid disease'
```

UNION ALL

```
SELECT 'Thyroid' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Others" = '1' AND  
"Any_Chronic_Illnes_/If_Other___specify" = 'Thyroid'
```

UNION ALL

```
SELECT 'Migraine' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Others" = '1' AND  
"Any_Chronic_Illnes_/If_Other___specify" = 'Migraine'
```

UNION ALL

-- Diseases from Household_members table

```
SELECT 'Tuberculosis' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Tuberculosis" = '1'
```

UNION ALL

```

SELECT 'Arthritis/Back Pain' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Arthritis/Back_Pain" = '1'

UNION ALL

SELECT 'Diabetes' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Diabetes" = '1'

UNION ALL

SELECT 'Paralysis' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Paralysis" = '1'

UNION ALL

SELECT 'High BP' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/High_BP" = '1'

UNION ALL

SELECT 'Heart Disease and stroke' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Heart_Disease_and_stroke" = '1'

UNION ALL

SELECT 'Cancer' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Cancer" = '1'

UNION ALL

SELECT 'Alzheimer's Disease' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Alzheimer's_Disease" = '1'

UNION ALL

SELECT 'Kidney Diseases' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Kidney_Diseases" =
'1'

UNION ALL

SELECT 'Liver Cirrhosis' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Liver_Cirrhosis" = '1'

UNION ALL

SELECT 'Peptic Ulcer (stomach)' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Peptic_Ulcer(stomach)" = '1'

UNION ALL

SELECT 'Alcohol related health issues' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Alcohol_related_health_issues" = '1'

UNION ALL

SELECT 'Smoking related health issues' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Smoking_related_health_issues" = '1'

UNION ALL

SELECT 'Asthma' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Asthma" = '1'

UNION ALL

SELECT 'Stomach operation' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Others" = '1' AND
"Any_Chronic_Illness_If_Other_specify" = 'Stomach_operation'

UNION ALL

SELECT 'Heartburn from oily food' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Others" =
'1' AND "Any_Chronic_Illness_If_Other_specify" = 'Heartburn_from_oily_food'

)

SELECT

Disease,

SUM(Count) AS TotalCount,

ROUND(CAST(SUM(Count) AS REAL) * 100 / (SELECT SUM(Count) FROM AllDiseases), 2) AS Percentage

```

FROM AllDiseases

GROUP BY Disease

HAVING TotalCount > 0

ORDER BY TotalCount DESC;

```Error: (sqlite3.ProgrammingError) You can only execute one statement at a time.

[SQL: WITH AllDiseases AS (

-- Diseases from Respondent table

SELECT 'Tuberculosis' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Tuberculosis" = '1'

UNION ALL

SELECT 'Arthritis/Back Pain' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Arthriti\_/Back\_Pain" = '1'

UNION ALL

SELECT 'Diabetes' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Diabetes" = '1'

UNION ALL

SELECT 'Paralysis' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Paralysis" = '1'

UNION ALL

SELECT 'High BP' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/High\_BP" = '1'

UNION ALL

SELECT 'Heart Disease and stroke' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE  
"Any\_Chronic\_Illnes\_/Heart\_Disease\_and\_stroke" = '1'

UNION ALL

SELECT 'Cancer' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Cancer" = '1'

UNION ALL

SELECT 'Alzheimer's Disease' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Alzheimer's\_Disease" =  
'1'

UNION ALL

SELECT 'Kidney Diseases' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Kidney\_Diseases" = '1'

UNION ALL

SELECT 'Liver Cirrhosis' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Liver\_Cirrhosis" = '1'

UNION ALL

SELECT 'Peptic Ulcer (stomach)' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE  
"Any\_Chronic\_Illnes\_/Peptic\_Ulcer(stomach)" = '1'

UNION ALL

SELECT 'Alcohol related health issues' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE  
"Any\_Chronic\_Illnes\_/Alcohol\_related\_health\_issues" = '1'

UNION ALL

SELECT 'Smoking related health issues' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE  
"Any\_Chronic\_Illnes\_/Smoking\_related\_health\_issues" = '1'

UNION ALL

SELECT 'Asthma' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Asthma" = '1'

UNION ALL

SELECT 'Low BP' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illnes\_/Others" = '1' AND  
"Any\_Chronic\_Illnes\_/If\_Other\_\_\_specify" = 'Low BP'

UNION ALL

```
SELECT 'Body issues' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Others" = '1' AND "Any_Chronic_Illnes_/If_Other___specify" = 'Body issues'
```

UNION ALL

```
SELECT 'Thyroid disease' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Others" = '1' AND "Any_Chronic_Illnes_/If_Other___specify" = 'Thyroid disease'
```

UNION ALL

```
SELECT 'Thyroid' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Others" = '1' AND "Any_Chronic_Illnes_/If_Other___specify" = 'Thyroid'
```

UNION ALL

```
SELECT 'Migraine' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Others" = '1' AND "Any_Chronic_Illnes_/If_Other___specify" = 'Migraine'
```

UNION ALL

-- Diseases from Household\_members table

```
SELECT 'Tuberculosis' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Tuberculosis" = '1'
```

UNION ALL

```
SELECT 'Arthritis/Back Pain' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Arthritis/Back_Pain" = '1'
```

UNION ALL

```
SELECT 'Diabetes' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Diabetes" = '1'
```

UNION ALL

```
SELECT 'Paralysis' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Paralysis" = '1'
```

UNION ALL

```
SELECT 'High BP' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/High_BP" = '1'
```

UNION ALL

```
SELECT 'Heart Disease and stroke' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Heart_Disease_and_stroke" = '1'
```

UNION ALL

```
SELECT 'Cancer' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Cancer" = '1'
```

UNION ALL

```
SELECT 'Alzheimer's Disease' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Alzheimer's_Disease" = '1'
```

UNION ALL

```
SELECT 'Kidney Diseases' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Kidney_Diseases" = '1'
```

UNION ALL

```
SELECT 'Liver Cirrhosis' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Liver_Cirrhosis" = '1'
```

UNION ALL

```
SELECT 'Peptic Ulcer (stomach)' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Peptic_Ulcer(stomach)" = '1'
```

UNION ALL

```

SELECT 'Alcohol related health issues' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Alcohol_related_health_issues" = '1'

UNION ALL

SELECT 'Smoking related health issues' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Smoking_related_health_issues" = '1'

UNION ALL

SELECT 'Asthma' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Asthma" = '1'

UNION ALL

SELECT 'Stomach operation' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Others" = '1' AND
"Any_Chronic_Illness>If_Other_specify," = 'Stomach_operation'

UNION ALL

SELECT 'Heartburn from oily food' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Others" =
'1' AND "Any_Chronic_Illness>If_Other_specify," = 'Heartburn_from_oily_food'
)
SELECT
Disease,
SUM(Count) AS TotalCount,
ROUND(CAST(SUM(Count) AS REAL) * 100 / (SELECT SUM(Count) FROM AllDiseases), 2) AS Percentage
FROM AllDiseases
GROUP BY Disease
HAVING TotalCount > 0
ORDER BY TotalCount DESC;
'''

```

(Background on this error at: <https://sqlalche.me/e/20/f405>) The error "You can only execute one statement at a time" indicates that there might be an extra semicolon or an unintended separation of queries. I've reviewed the query and it appears to be a single, well-formed CTE followed by a SELECT statement.

The previous 'Action Input' did not have the extra backticks, so that was not the issue. The error message itself is a bit misleading given the structure of the query.

However, I notice that the 'sql\_db\_query\_checker' output also included the '```sqlite` and '``` ` markers, which might be causing the issue when the 'sql\_db\_query' tool tries to execute it. I will try to execute the query again, ensuring that *only* the SQL query text is passed to the 'sql\_db\_query' tool, without any markdown formatting.

Also, I will simplify the 'WHERE' clause for the 'ROUND' function in the main 'SELECT' statement. Instead of 'WHERE Disease IS NOT NULL AND TotalCount > 0', I will use 'HAVING TotalCount > 0' as it's more appropriate for filtering aggregated results. This was already done in the 'sql\_db\_query\_checker' step, so the query itself should be fine. The issue is likely with the tool execution environment.

Let's try executing the query directly without any markdown formatting in the 'Action Input'.

Action: sql\_db\_query

Action Input: WITH AllDiseases AS (

-- Diseases from Respondent table

SELECT 'Tuberculosis' AS Disease, COUNT(\*) AS Count FROM "Respondent" WHERE "Any\_Chronic\_Illness/Tuberculosis" = '1'

UNION ALL

```

SELECT 'Arthritis/Back Pain' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Arthriti_/Back_Pain" = '1'

UNION ALL

SELECT 'Diabetes' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Diabetes" = '1'

UNION ALL

SELECT 'Paralysis' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Paralysis" = '1'

UNION ALL

SELECT 'High BP' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/High_BP" = '1'

UNION ALL

SELECT 'Heart Disease and stroke' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE
"Any_Chronic_Illnes_/Heart_Disease_and_stroke" = '1'

UNION ALL

SELECT 'Cancer' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Cancer" = '1'

UNION ALL

SELECT 'Alzheimer's Disease' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Alzheimer's_Disease" =
'1'

UNION ALL

SELECT 'Kidney Diseases' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Kidney_Diseases" = '1'

UNION ALL

SELECT 'Liver Cirrhosis' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Liver_Cirrhosis" = '1'

UNION ALL

SELECT 'Peptic Ulcer (stomach)' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE
"Any_Chronic_Illnes_/Peptic_Ulcer(stomach)" = '1'

UNION ALL

SELECT 'Alcohol related health issues' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE
"Any_Chronic_Illnes_/Alcohol_related_health_issues" = '1'

UNION ALL

SELECT 'Smoking related health issues' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE
"Any_Chronic_Illnes_/Smoking_related_health_issues" = '1'

UNION ALL

SELECT 'Asthma' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Asthma" = '1'

UNION ALL

SELECT 'Low BP' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Others" = '1' AND
"Any_Chronic_Illnes_/If_Other___specify" = 'Low BP'

UNION ALL

SELECT 'Body issues' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Others" = '1' AND
"Any_Chronic_Illnes_/If_Other___specify" = 'Body issues'

UNION ALL

SELECT 'Thyroid disease' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Others" = '1' AND
"Any_Chronic_Illnes_/If_Other___specify" = 'Thyroid disease'

UNION ALL

SELECT 'Thyroid' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Others" = '1' AND
"Any_Chronic_Illnes_/If_Other___specify" = 'Thyroid'

UNION ALL

```

```
SELECT 'Migraine' AS Disease, COUNT(*) AS Count FROM "Respondent" WHERE "Any_Chronic_Illnes_/Others" = '1' AND
"Any_Chronic_Illnes_/If_Other_specify" = 'Migraine'
```

UNION ALL

-- Diseases from Household\_members table

```
SELECT 'Tuberculosis' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Tuberculosis" = '1'
```

UNION ALL

```
SELECT 'Arthritis/Back Pain' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Arthritis/Back_Pain" = '1'
```

UNION ALL

```
SELECT 'Diabetes' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Diabetes" = '1'
```

UNION ALL

```
SELECT 'Paralysis' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Paralysis" = '1'
```

UNION ALL

```
SELECT 'High BP' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/High_BP" = '1'
```

UNION ALL

```
SELECT 'Heart Disease and stroke' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Heart_Disease_and_stroke" = '1'
```

UNION ALL

```
SELECT 'Cancer' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Cancer" = '1'
```

UNION ALL

```
SELECT 'Alzheimer's Disease' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Alzheimer's_Disease" = '1'
```

UNION ALL

```
SELECT 'Kidney Diseases' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Kidney_Diseases" =
'1'
```

UNION ALL

```
SELECT 'Liver Cirrhosis' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Liver_Cirrhosis" = '1'
```

UNION ALL

```
SELECT 'Peptic Ulcer (stomach)' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Peptic_Ulcer(stomach)" = '1'
```

UNION ALL

```
SELECT 'Alcohol related health issues' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Alcohol_related_health_issues" = '1'
```

UNION ALL

```
SELECT 'Smoking related health issues' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE
"Any_Chronic_Illness/Smoking_related_health_issues" = '1'
```

UNION ALL

```
SELECT 'Asthma' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Asthma" = '1'
```

UNION ALL

```
SELECT 'Stomach operation' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE "Any_Chronic_Illness/Others" = '1' AND
"Any_Chronic_Illness_/If_Other_specify;" = 'Stomach_operation'
```

UNION ALL

```
SELECT 'Heartburn from oily food' AS Disease, COUNT(*) AS Count FROM "Household_members" WHERE 'Any_Chronic_Illness/Others' = '1' AND 'Any_Chronic_Illness_If_Other_specify' = 'Heartburn_from_oily_food'
```

)

SELECT

Disease,

SUM(Count) AS TotalCount,

ROUND(CAST(SUM(Count) AS REAL) \* 100 / (SELECT SUM(Count) FROM AllDiseases), 2) AS Percentage

FROM AllDiseases

GROUP BY Disease

HAVING TotalCount > 0

ORDER BY TotalCount DESC;(['Arthritis/Back Pain', 9, 25.0), ('High BP', 8, 22.22), ('Diabetes', 6, 16.67), ('Low BP', 3, 8.33), ('Thyroid disease', 1, 2.78), ('Thyroid', 1, 2.78), ('Stomach operation', 1, 2.78), ('Smoking related health issues', 1, 2.78), ('Migraine', 1, 2.78), ('Liver Cirrhosis', 1, 2.78), ('Heartburn from oily food', 1, 2.78), ('Body issues', 1, 2.78), ('Asthma', 1, 2.78), ('Alcohol related health issues', 1, 2.78)])

The composition of chronic diseases among respondents and their families is as follows:

- \* **Arthritis/Back Pain**: 9 occurrences (25.0%)
- \* **High BP**: 8 occurrences (22.22%)
- \* **Diabetes**: 6 occurrences (16.67%)
- \* **Low BP**: 3 occurrences (8.33%)
- \* **Thyroid disease**: 1 occurrence (2.78%)
- \* **Thyroid**: 1 occurrence (2.78%)
- \* **Stomach operation**: 1 occurrence (2.78%)
- \* **Smoking related health issues**: 1 occurrence (2.78%)
- \* **Migraine**: 1 occurrence (2.78%)
- \* **Liver Cirrhosis**: 1 occurrence (2.78%)
- \* **Heartburn from oily food**: 1 occurrence (2.78%)
- \* **Body issues**: 1 occurrence (2.78%)
- \* **Asthma**: 1 occurrence (2.78%)
- \* **Alcohol related health issues**: 1 occurrence (2.78%)

Final Answer: The most prevalent chronic diseases are Arthritis/Back Pain (25.0%), High BP (22.22%), and Diabetes (16.67%). Other reported illnesses include Low BP (8.33%), and several conditions like Thyroid disease, Thyroid, Stomach operation, Smoking related health issues, Migraine, Liver Cirrhosis, Heartburn from oily food, Body issues, Asthma, and Alcohol related health issues, each accounting for 2.78% of the total.

> Finished chain.

=====

=====

The most prevalent chronic diseases are Arthritis/Back Pain (25.0%), High BP (22.22%), and Diabetes (16.67%). Other reported illnesses include Low BP (8.33%), and several conditions like Thyroid disease, Thyroid, Stomach operation, Smoking related health issues, Migraine, Liver Cirrhosis, Heartburn from oily food, Body issues, Asthma, and Alcohol related health issues, each accounting for 2.78% of the total.

Query (or 'exit'): exit

Session ended. Goodbye!

## 7. What are the composition of sources of drinking water, type of electricity connection, frequency of waste collection, and toilets connected to a sewer line?

--- SQL Agent Interactive Mode ---

Type your database question and press Enter. Type 'exit' or 'quit' to end the session.

Query (or 'exit'): What are the composition of toilets connected to a sewer line?

--- Processing Query: What are the composition of toilets connected to a sewer line? ---

> Entering new SQL Agent Executor chain...

Action: *sql\_db\_query\_checker*

Action Input: *SELECT "Do\_you\_have\_toilets\_in\_your\_house\_", "Is\_your\_house\_connected\_to\_the\_sewer\_system\_" FROM "Respondent";SELECT "Do\_you\_have\_toilets\_in\_your\_house\_", "Is\_your\_house\_connected\_to\_the\_sewer\_system\_" FROM "Respondent";The user is asking for the composition of toilets connected to a sewer line. This means I need to count how many houses have toilets and are connected to a sewer system, and also how many have toilets but are not connected, or don't have toilets at all.*

*I will query the `Respondent` table for the columns `Do\_you\_have\_toilets\_in\_your\_house\_` and `Is\_your\_house\_connected\_to\_the\_sewer\_system\_`. Then I will group the results by these two columns and count the occurrences.*

Action: *sql\_db\_query*

Action Input: *SELECT "Do\_you\_have\_toilets\_in\_your\_house\_", "Is\_your\_house\_connected\_to\_the\_sewer\_system\_", COUNT(\*) FROM "Respondent" GROUP BY "Do\_you\_have\_toilets\_in\_your\_house\_", "Is\_your\_house\_connected\_to\_the\_sewer\_system\_";[(, ", 1), ('Yes', 'No,*

1), ('Yes', 'Yes', 27)]Final Answer: Out of the respondents, 27 houses have toilets connected to a sewer system, 1 house has toilets but is not connected to a sewer system, and 1 respondent did not provide information regarding toilets or sewer connection.

> Finished chain.

=====  
=====

Out of the respondents, 27 houses have toilets connected to a sewer system, 1 house has toilets but is not connected to a sewer system, and 1 respondent did not provide information regarding toilets or sewer connection.

Query (or 'exit'): exit

Session ended. Goodbye!

**8. What is the composition of early warnings received by type of early warning (flood, extreme heat, etc.) and the mode of receiving such warnings.**

## **9. What is the composition of cooling appliances owned by the respondents? Categorize by religion and caste.**

--- SQL Agent Interactive Mode ---

Type your database question and press Enter. Type 'exit' or 'quit' to end the session.

Query (or 'exit'): What is the composition of cooling appliances owned by the respondents? Categorize by religion and caste.

--- Processing Query: What is the composition of cooling appliances owned by the respondents? Categorize by religion and caste. ---

> Entering new SQL Agent Executor chain...

**Action: sql\_db\_query\_checker**

**Action Input: SELECT**

**"Religion",**

**"Category\_of\_Caste",**

```

SUM(CASE WHEN
"What_kinds_of_cooling_appliances_an_/or_cooling_techniques_do_you_currently
_have_in_your_house_/Fan" = '1' THEN 1 ELSE 0 END) AS "Fan_Count",

SUM(CASE WHEN
"What_kinds_of_cooling_appliances_an_/or_cooling_techniques_do_you_currently
_have_in_your_house_/Air_cooler" = '1' THEN 1 ELSE 0 END) AS "Air_Cooler_Count",

SUM(CASE WHEN
"What_kinds_of_cooling_appliances_an_/or_cooling_techniques_do_you_currently
_have_in_your_house_/Air_conditioner" = '1' THEN 1 ELSE 0 END) AS
"Air_Conditioner_Count",

SUM(CASE WHEN
"What_kinds_of_cooling_appliances_an_/or_cooling_techniques_do_you_currently
_have_in_your_house_/Fridge" = '1' THEN 1 ELSE 0 END) AS "Fridge_Count",

SUM(CASE WHEN
"What_kinds_of_cooling_appliances_an_/or_cooling_techniques_do_you_currently
_have_in_your_house_/Cooling_paint" = '1' THEN 1 ELSE 0 END) AS
"Cooling_Paint_Count",

SUM(CASE WHEN
"What_kinds_of_cooling_appliances_an_/or_cooling_techniques_do_you_currently
_have_in_your_house_/False_ceiling" = '1' THEN 1 ELSE 0 END) AS
"False_Ceiling_Count",

SUM(CASE WHEN
"What_kinds_of_cooling_appliances_an_/or_cooling_techniques_do_you_currently
_have_in_your_house_/Green_cover" = '1' THEN 1 ELSE 0 END) AS
"Green_Cover_Count",

SUM(CASE WHEN
"What_kinds_of_cooling_appliances_an_/or_cooling_techniques_do_you_currently
_have_in_your_house_/Additional_ventilation" = '1' THEN 1 ELSE 0 END) AS
"Additional_Ventilation_Count",

SUM(CASE WHEN
"What_kinds_of_cooling_appliances_an_/or_cooling_techniques_do_you_currently
_have_in_your_house_/Others" = '1' THEN 1 ELSE 0 END) AS
"Others_Cooling_Count"

FROM "Respondent"

GROUP BY "Religion", "Category_of_Caste"

ORDER BY "Religion", "Category_of_Caste";`` ` sql

```

**SELECT**

**"Religion",**

**"Category\_of\_Caste",**

**SUM(CASE WHEN**

**"What\_kinds\_of\_cooling\_appliances\_an\_/or\_cooling\_techniques\_do\_you\_currently  
\_have\_in\_your\_house\_/Fan" = '1' THEN 1 ELSE 0 END) AS "Fan\_Count",**

**SUM(CASE WHEN**

**"What\_kinds\_of\_cooling\_appliances\_an\_/or\_cooling\_techniques\_do\_you\_currently  
\_have\_in\_your\_house\_/Air\_cooler" = '1' THEN 1 ELSE 0 END) AS "Air\_Cooler\_Count",**

**SUM(CASE WHEN**

**"What\_kinds\_of\_cooling\_appliances\_an\_/or\_cooling\_techniques\_do\_you\_currently  
\_have\_in\_your\_house\_/Air\_conditioner" = '1' THEN 1 ELSE 0 END) AS  
"Air\_Cooler\_Count",**

**SUM(CASE WHEN**

**"What\_kinds\_of\_cooling\_appliances\_an\_/or\_cooling\_techniques\_do\_you\_currently  
\_have\_in\_your\_house\_/Fridge" = '1' THEN 1 ELSE 0 END) AS "Fridge\_Count",**

**SUM(CASE WHEN**

**"What\_kinds\_of\_cooling\_appliances\_an\_/or\_cooling\_techniques\_do\_you\_currently  
\_have\_in\_your\_house\_/Cooling\_paint" = '1' THEN 1 ELSE 0 END) AS  
"Cooling\_Paint\_Count",**

**SUM(CASE WHEN**

**"What\_kinds\_of\_cooling\_appliances\_an\_/or\_cooling\_techniques\_do\_you\_currently  
\_have\_in\_your\_house\_/False\_ceiling" = '1' THEN 1 ELSE 0 END) AS  
"False\_Ceiling\_Count",**

**SUM(CASE WHEN**

**"What\_kinds\_of\_cooling\_appliances\_an\_/or\_cooling\_techniques\_do\_you\_currently  
\_have\_in\_your\_house\_/Green\_cover" = '1' THEN 1 ELSE 0 END) AS  
"Green\_Cover\_Count",**

**SUM(CASE WHEN**

**"What\_kinds\_of\_cooling\_appliances\_an\_/or\_cooling\_techniques\_do\_you\_currently  
\_have\_in\_your\_house\_/Additional\_ventilation" = '1' THEN 1 ELSE 0 END) AS  
"Additional\_Ventilation\_Count",**

**SUM(CASE WHEN**

**"What\_kinds\_of\_cooling\_appliances\_an\_/or\_cooling\_techniques\_do\_you\_currently  
\_have\_in\_your\_house\_/Others" = '1' THEN 1 ELSE 0 END) AS  
"Others\_Cooling\_Count"**

**FROM "Respondent"**

**GROUP BY "Religion", "Category\_of\_Caste"**

**ORDER BY "Religion", "Category\_of\_Caste";**

**``` Action: sql\_db\_query**

**Action Input: SELECT**

**"Religion",**

**"Category\_of\_Caste",**

**SUM(CASE WHEN**

**"What\_kinds\_of\_cooling\_appliances\_an\_/or\_cooling\_techniques\_do\_you\_currently  
\_have\_in\_your\_house\_/Fan" = '1' THEN 1 ELSE 0 END) AS "Fan\_Count",**

**SUM(CASE WHEN**

**"What\_kinds\_of\_cooling\_appliances\_an\_/or\_cooling\_techniques\_do\_you\_currently  
\_have\_in\_your\_house\_/Air\_cooler" = '1' THEN 1 ELSE 0 END) AS "Air\_Cooler\_Count",**

**SUM(CASE WHEN**

**"What\_kinds\_of\_cooling\_appliances\_an\_/or\_cooling\_techniques\_do\_you\_currently  
\_have\_in\_your\_house\_/Air\_conditioner" = '1' THEN 1 ELSE 0 END) AS  
"Air\_Conditioner\_Count",**

**SUM(CASE WHEN**

**"What\_kinds\_of\_cooling\_appliances\_an\_/or\_cooling\_techniques\_do\_you\_currently  
\_have\_in\_your\_house\_/Fridge" = '1' THEN 1 ELSE 0 END) AS "Fridge\_Count",**

**SUM(CASE WHEN**

**"What\_kinds\_of\_cooling\_appliances\_an\_/or\_cooling\_techniques\_do\_you\_currently  
\_have\_in\_your\_house\_/Cooling\_paint" = '1' THEN 1 ELSE 0 END) AS  
"Cooling\_Paint\_Count",**

**SUM(CASE WHEN**

**"What\_kinds\_of\_cooling\_appliances\_an\_/or\_cooling\_techniques\_do\_you\_currently  
\_have\_in\_your\_house\_/False\_ceiling" = '1' THEN 1 ELSE 0 END) AS  
"False\_Ceiling\_Count",**

**SUM(CASE WHEN**

**"What\_kinds\_of\_cooling\_appliances\_an\_/or\_cooling\_techniques\_do\_you\_currently  
\_have\_in\_your\_house\_/Green\_cover" = '1' THEN 1 ELSE 0 END) AS  
"Green\_Cover\_Count",**

**SUM(CASE WHEN**

**"What\_kinds\_of\_cooling\_appliances\_an\_/or\_cooling\_techniques\_do\_you\_currently**

```

_have_in_your_house_/Additional_ventilation" = '1' THEN 1 ELSE 0 END) AS
"Additional_Ventilation_Count",

SUM(CASE WHEN
"What_kinds_of_cooling_appliances_an_/or_cooling_techniques_do_you_currently
_have_in_your_house_/Others" = '1' THEN 1 ELSE 0 END) AS
"Others_Cooling_Count"

FROM "Respondent"

GROUP BY "Religion", "Category_of_Caste"

ORDER BY "Religion", "Category_of_Caste";[(, 'OBC', 1, 0, 0, 1, 0, 0, 0, 1, 0), (, 'SC', 1,
1, 0, 1, 0, 0, 0, 0, 0), ('Christian', 'OBC', 2, 0, 0, 2, 0, 0, 0, 0, 0), ('Christian', 'SC', 1, 1, 0,
1, 0, 0, 0, 0, 0), ('Hindu', ' ', 2, 0, 0, 1, 0, 0, 0, 0, 0), ('Hindu', 'General', 1, 1, 0, 1, 0, 0, 0, 0,
0), ('Hindu', 'OBC', 4, 4, 0, 2, 0, 0, 0, 1, 0), ('Hindu', 'SC', 3, 0, 0, 3, 0, 0, 0, 0, 0), ('Hindu',
'ST', 6, 4, 0, 4, 1, 0, 1, 0, 0), ('Muslim', ' ', 2, 1, 0, 2, 0, 0, 0, 0, 0), ('Muslim', 'General', 1, 1,
0, 1, 0, 0, 0, 0, 0), ('Prefer Not to Say', 'SC', 1, 1, 0, 0, 0, 0, 0, 0, 0), ('Prefer Not to Say',
'ST', 1, 0, 0, 1, 0, 0, 0, 0, 0)]Final Answer: The composition of cooling appliances
owned by respondents, categorized by religion and caste, is as follows:

```

- \* **\*\*Religion: (Empty/Not Specified)\*\***
  - \* **\*\*Caste: OBC\*\* - 1 Fan, 1 Fridge, 1 Additional Ventilation.**
  - \* **\*\*Caste: SC\*\* - 1 Fan, 1 Air Cooler, 1 Fridge.**
- \* **\*\*Religion: Christian\*\***
  - \* **\*\*Caste: OBC\*\* - 2 Fans, 2 Fridges.**
  - \* **\*\*Caste: SC\*\* - 1 Fan, 1 Air Cooler, 1 Fridge.**
- \* **\*\*Religion: Hindu\*\***
  - \* **\*\*Caste: (Empty/Not Specified)\*\* - 2 Fans, 1 Fridge.**
  - \* **\*\*Caste: General\*\* - 1 Fan, 1 Air Cooler, 1 Fridge.**
  - \* **\*\*Caste: OBC\*\* - 4 Fans, 4 Air Coolers, 2 Fridges, 1 Additional Ventilation.**
  - \* **\*\*Caste: SC\*\* - 3 Fans, 3 Fridges.**
  - \* **\*\*Caste: ST\*\* - 6 Fans, 4 Air Coolers, 4 Fridges, 1 Cooling Paint, 1 Green Cover.**
- \* **\*\*Religion: Muslim\*\***
  - \* **\*\*Caste: (Empty/Not Specified)\*\* - 2 Fans, 1 Air Cooler, 2 Fridges.**

\* **\*\*Caste: General\*\* - 1 Fan, 1 Air Cooler, 1 Fridge.**

\* **\*\*Religion: Prefer Not to Say\*\***

\* **\*\*Caste: SC\*\* - 1 Fan, 1 Air Cooler.**

\* **\*\*Caste: ST\*\* - 1 Fan, 1 Fridge.**

> **Finished chain.**

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The composition of cooling appliances owned by respondents, categorized by religion and caste, is as follows:

\* **\*\*Religion: (Empty/Not Specified)\*\***

\* **\*\*Caste: OBC\*\* - 1 Fan, 1 Fridge, 1 Additional Ventilation.**

\* **\*\*Caste: SC\*\* - 1 Fan, 1 Air Cooler, 1 Fridge.**

\* **\*\*Religion: Christian\*\***

\* **\*\*Caste: OBC\*\* - 2 Fans, 2 Fridges.**

\* **\*\*Caste: SC\*\* - 1 Fan, 1 Air Cooler, 1 Fridge.**

\* **\*\*Religion: Hindu\*\***

\* **\*\*Caste: (Empty/Not Specified)\*\* - 2 Fans, 1 Fridge.**

\* **\*\*Caste: General\*\* - 1 Fan, 1 Air Cooler, 1 Fridge.**

\* **\*\*Caste: OBC\*\* - 4 Fans, 4 Air Coolers, 2 Fridges, 1 Additional Ventilation.**

\* **\*\*Caste: SC\*\* - 3 Fans, 3 Fridges.**

\* **\*\*Caste: ST\*\* - 6 Fans, 4 Air Coolers, 4 Fridges, 1 Cooling Paint, 1 Green Cover.**

\* **\*\*Religion: Muslim\*\***

\* **\*\*Caste: (Empty/Not Specified)\*\* - 2 Fans, 1 Air Cooler, 2 Fridges.**

\* **\*\*Caste: General\*\* - 1 Fan, 1 Air Cooler, 1 Fridge.**

\* **\*\*Religion: Prefer Not to Say\*\***

\* \*\*Caste: SC\*\* - 1 Fan, 1 Air Cooler.

\* \*\*Caste: ST\*\* - 1 Fan, 1 Fridge.

Query (or 'exit'): exit

Session ended. Goodbye!